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Chain collapse, reduced dielectric, and water release drive protein phase separation

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ABSTRACT Biomolecular condensates represent unique microenvironments that organize intracellular biology and promote biochemical reactions. However, the biomolecular interactions driving condensate phase separation are often weak, transient, and heterogeneous. Investigating the structural biology and chemical properties of condensate interiors has therefore proven experimentally challenging, often requiring the use of perturbative probes. To overcome this challenge, we combine label-free optical scattering and vibrational spectroscopy approaches spanning ultraviolet, visible, mid-infrared, and terahertz wavelengths with deep-learning-based ensemble prediction of intrinsically disordered protein conformations. This suite of label-free approaches provides quantitative insights into protein-protein/protein-solvent interactions and the chemical properties of condensate interiors. Investigating the N-terminal domain of the RNA DEAD-box helicase 4, our experimental and computational results support a model of phase separation involving protein chain collapse, reduced dielectric, and water release. These molecular events are expected to enhance the strength of multivalent protein-protein interactions within condensates, creating a positive-feedback loop important for condensate growth and phase separation.

SIGNIFICANCE Biomolecular condensates organize cellular matter and play crucial roles in cell signaling by creating distinct environments to control molecular interactions and biochemical reactions. However, most experimental studies performed under ambient conditions use labels that perturb molecular interactions. Here, we use label-free vibrational spectroscopies and optical scattering to characterize protein structures. Combined with deep-learning methods, we quantitatively analyze changes in protein size, solvent dielectric, and water content during phase separation. Our findings reveal chain collapse, reduced dielectric, and water release are involved in phase separation. The reduction in dielectric and release of water could strengthen multivalent protein-protein interactions that are essential for condensation. Our work offers mechanistic insights into a positive-feedback loop that can be generally important for liquid-liquid phase separation.

INTRODUCTION

Over the last decade, studies have revealed that biomolecular condensates play crucial roles in the organization of intracellular biology and cell signaling by modulating molecular interactions and biochemical reactions.[1–7] Biomolecular condensates are driven by multivalent interactions, such as cation-π interactions, π-π stacking, hydrogen bonding, electrostatics, and dipole-dipole interactions.[1,8–11] Although such molecular interactions have been well characterized in dilute conditions, how the internal environment of a condensate affects these interactions remains challenging to investigate.[12–15] Most previous experimental studies of biological condensates performed under ambient

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conditions (i.e., typical room temperatures and humidity at 1 atmospheric pressure) utilized labels, such as fluorescent probes.[16] However, these labels can skew the phase separation process because phase separation is driven by weak, transient, and heterogeneous multivalent molecular interactions. Hence, the free energy landscapes of biological phase separation are susceptible to perturbation by labels.[17–19] To better address mechanistic questions, the field will benefit from the implementation and development of label-free biophysical methods.[20–24]

Here, we apply a suite of label-free optical scattering and vibrational spectroscopic methods to study protein liquid-liquid phase separation (Figure 1) of the intrinsically disordered N-terminal domain (NTD) consisting of the first 245 amino acids of the RNA DEAD-box helicase 4 (DDX4$^{[1\text{-}245]}$).[25–28] UV resonance Raman spectroscopy and confocal Raman microscopy characterize protein secondary structures of DDX4$^{[1\text{-}245]}$. Dynamic light scattering monitors the changes in size upon phase separation. Terahertz spectroscopy measures the dielectric constant in the DDX4$^{[1\text{-}245]}$ condensed phase. Finally, stimulated Raman imaging quantifies water content in the interior and exterior of the DDX4$^{[1\text{-}245]}$ condensates. Our results lead us to a model of DDX4$^{[1\text{-}245]}$ phase separation involving protein chain collapse, reduction of the local dielectric constant, and release of water molecules. The molecular events of dielectric reduction and water release are particularly important, as they can cooperatively strengthen the multivalent protein-protein interactions that are essential for condensation. Thus, our work reveals the possibility of a positive-feedback loop potentially key to the liquid-liquid phase separation.

MATERIALS AND METHODS

Expression and purification of DDX4$^{[1\text{-}245]}$

DDX4$^{[1\text{-}245]}$ (see supplemental information for amino acid sequence) was recombinantly expressed in *E. coli* with an N-terminal GST domain and TEV-protease cleavage site. From glycerol stock, bacteria inoculates were grown in LB containing 0.1 mg/mL ampicillin at 37°C overnight. The inoculates were transferred to 1 L of LB with ampicillin and grown at 37°C until $OD_{600} = 0.6–0.8$ AU. The cultures were then induced with 0.25 mM IPTG. The cultures were immediately brought to room temperature and grown overnight. Cells were collected by centrifugation and

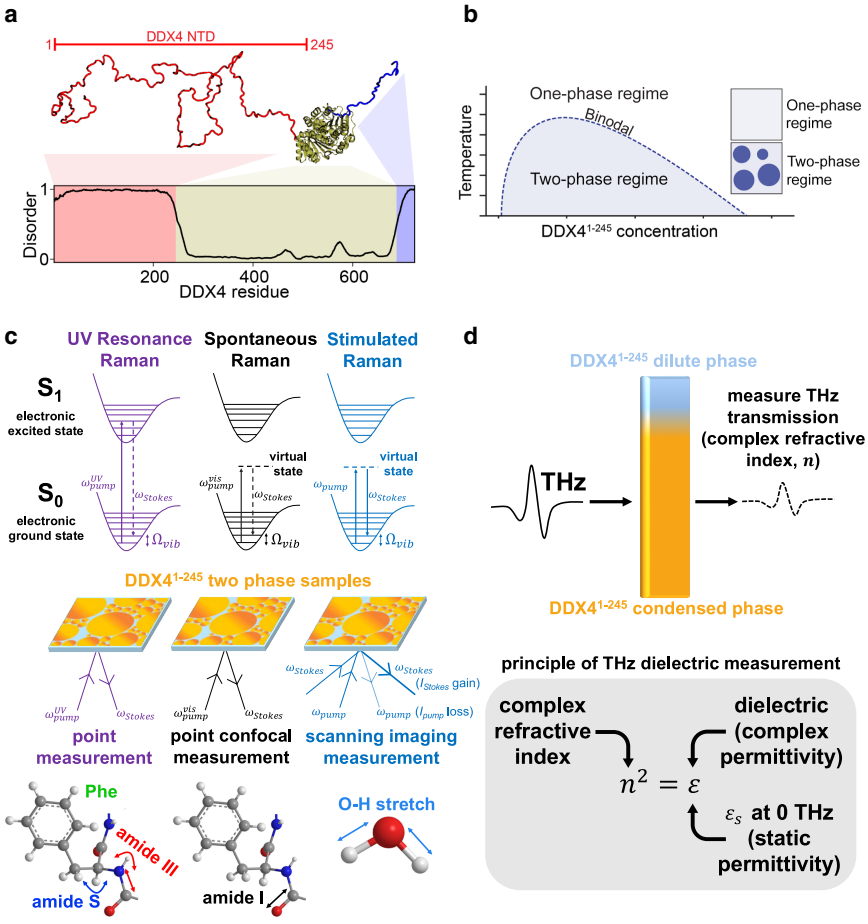


Figure 1. Vibrational spectroscopies to probe liquid-liquid phase separation of DDX4$^{[1\text{-}245]}$. (a) Representative computational structural conformation and disorder of full-length DDX4, including the DDX4$^{[1\text{-}245]}$ N-terminal domain (NTD) studied here. (b) An idealized phase diagram of DDX4$^{[1\text{-}245]}$ phase separation. (c) Simplified Jablonski diagrams and observables for UV resonance Raman, spontaneous confocal Raman, and stimulated Raman imaging. UV resonance Raman point measurements detect DDX4$^{[1\text{-}245]}$ phenylalanine (Phe) protein side-chain vibrations and determine the structure of DDX4$^{[1\text{-}245]}$ by probing the amide S (protein backbone C_α-H bending) and amide III vibrations (coupled protein backbone C-N stretching and N-H bending). Spontaneous Raman confocal imaging determines the structure of DDX4$^{[1\text{-}245]}$ inside individual condensates by probing the protein amide I vibration (mainly protein backbone C=O stretching). Stimulated Raman imaging quantifies the water content inside individual DDX4$^{[1\text{-}245]}$ condensates and at the condensate surface by probing the water O-H stretching vibration. (d) Terahertz (THz) spectroscopy measures the frequency-dependent complex refractive index (n) of the sample to extract the complex static permittivity (i.e., dielectric constant at zero frequency, ε_S) inside the DDX4$^{[1\text{-}245]}$ condensed phase.

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resuspended in lysis buffer (50 mM Tris, 500 mM NaCl, 5 mM DTT, pH = 8) containing protease inhibitor cocktail (cOmplete Mini, EDTA-free, Sigma-Aldrich, Cat. No. Roche-11836170001). Cells were sonicated on ice three times for 7 min each time in steps of 3 s (followed by 5 s off). The sonicate was clarified by centrifugation.

GST beads (Glutathione Sepharose 4B, GE Life Sciences, Cat. No. 17-0756-01) were washed three times with 5 mL binding buffer (500 mM NaCl, 10 mM Na_2HPO_4, 1.8 mM KH_2PO_4, pH = 7.3) in 15-mL Falcon tubes (1 mL of beads per 1 L of cell culture, using 1 mL beads per Falcon tube). Beads were subsequently washed once in 5 mL lysis buffer. The buffer was removed, and approximately 12 mL of clarified sonicate was applied per 1 mL of GST beads. The Falcon tubes were nutated overnight at 4°C to ensure complete binding of protein to beads.

Beads were decanted by centrifugation at 4°C and washed twice with 5 mL lysis buffer per 1 mL beads. TEV protease solution (20 μL ProTEV Plus, Promega, Cat. No. V6101, in 980 μL lysis buffer) was applied to 1 mL of GST-TEV-DDX4$^{[1\text{-}245]}$-bound beads. Beads were incubated at room temperature for 5 h while nutating. The supernatant containing TEV protease and DDX4$^{[1\text{-}245]}$ was decanted by centrifugation at room temperature. Beads were washed with 1 mL lysis buffer, centrifuged, and the supernatant collected to recover remaining protein.

FPLC purification (Superdex 75 prep grade, GE Life Sciences) was performed in storage buffer (500 mM NaCl, 20 mM Tris, 5 mM TCEP, pH 8 in H_2O) and used to separate TEV protease (48 kDa) from DDX4$^{[1\text{-}245]}$ (26.7 kDa). This yielded approximately 9–10 mg of DDX4$^{[1\text{-}245]}$ per liter of bacterial culture. Purified DDX4$^{[1\text{-}245]}$ was verified by SDS-PAGE, which showed a single band at ∼26 kDa (Figure S1). DDX4$^{[1\text{-}245]}$ was concentrated to 500–600 μM (Amicon Ultra, 3-kDa cutoff, Millipore, Cat. No. UFC900324). Aliquots were flash frozen in liquid nitrogen and stored at −80°C until use. Aliquots were thawed at room temperature for at least 30 min prior to use in all experiments.

UV resonance Raman spectroscopy

FPLC-purified DDX4$^{[1\text{-}245]}$ in storage buffer was quickly diluted ("quenched") at room temperature to a final concentration of 50 μM DDX4$^{[1\text{-}245]}$ and 125 mM NaCl (with 20 mM Tris, 5 mM TCEP, pH = 8 in H_2O). This condition yielded phase-separated condensates while avoiding the growth of larger condensates over the duration of the experiments. The sample was probed at room temperature (25°C) using a home-built UV resonance Raman spectrophotometer. Briefly, the 215-nm light output from the fourth harmonic of a 1-kHz Ti:Sapphire laser was focused on a vertically mounted, fused silica microcapillary with inner diameter of 100 μm. The protein sample was flowed at a rate to ensure fresh sample for each pulse. The Raman-scattered photons were collected, dispersed through a prism prefilter and spectrograph, and detected by a CCD. The power at the sample was less than 0.8 mW. Five 1-min spectra of protein sample and buffer were collected to generate the final UV resonance Raman difference spectra of protein.

Confocal Raman spectroscopy

FPLC-purified DDX4$^{[1\text{-}245]}$ was quenched at room temperature to a final concentration of 115 μM in 100 mM NaCl, 20 mM Tris, 5 mM TCEP (pH = 8) (pD = pH + 0.4 = 8) in H_2O (or D_2O), yielding phase-separated condensates. The phase-separated mixture was applied to a glass slide (Fisher Scientific, Cat. No. 12-544-4) and sealed with a glass cover slip (Fisher Scientific, Cat. No. 12-545-A) using a spacer (Sigma-Aldrich, Grace Bio-Labs SecureSeal Imaging Spacer, GBL654008). The sample was probed at room temperature using a home-built confocal Raman microscope. A 532-nm laser (Samba 532 nm, 400 mW, Cobolt Inc.) is used as the light source. The laser beam was first collimated and expanded by telescope lenses (f_1 = 100 mm, f_2 = 300 mm, Thorlabs) to achieve the designed illumination spot size. The expanded beam was then directed to an inverted microscope (IX71, Olympus) with a dichroic beam splitter installed (LPD1, LPD02-532RU-25, Semrock). A UplanSApo 20×, N.A. = 0.75 (Olympus) objective was used. The emitted Raman signal first passed a pinhole (100 μm, Thorlabs) for background suppression and was relayed by two lenses (Thorlabs) before being projected to the spectrometer (Kymera 328i, Andor) that is equipped with a grating of 600 lines/mm, blazed at 500 nm. A long-pass filter (LP03-532RU-25, Semrock) was installed between the relay lenses to block laser light from Rayleigh scattering. Raman signal was then collected by an EMCCD with an active pixel-area of 200 × 1,600 (Newton970, Andor). All the data collection was performed with a custom LABVIEW program (National Instruments). Using a laser power of 0.1 mW, spectra were collected by moving the laser beam inside or outside of condensates. No laser-induced changes of condensates were observed inside of condensates. When measuring outside of the condensates, small condensates could become optically trapped in the path of the laser, although this occurred on timescales longer than our experiments. Each reported spectrum is an average of 10 acquisitions, with each acquisition in H_2O or D_2O collected for an exposure time of 55 s or 40 s, respectively.

Dynamic light scattering

FPLC-purified DDX4$^{[1\text{-}245]}$ in storage buffer was filtered (GE Healthcare Life Sciences, Whatman Anotop 10-mm syringe filter, 0.1 μM pore size, Cat. No. 6809-1012) at room temperature and prepared at the concentrations of DDX4$^{[1\text{-}245]}$ and NaCl described in the main text (all conditions at 20 mM Tris, 5 mM TCEP, pH = 8). Experiments were carried out using a Wyatt DynaPro

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PlateReader II instrument. Sample solutions (50–100 μL) were transferred to wells in a 384-well glass-bottom plate (Greiner Bio-One SensoPlate, VWR Cat. No. 82051-546). For each experiment, 60–100 acquisitions were collected, 5 s per acquisition, and the quality of the autocorrelation function was verified. The instrument was cycled from 25°C to 10°C to 25°C, and measurements were performed on the same sample at these temperatures. Our dynamic light scattering (DLS) data were analyzed using the Dynamics software (Wyatt Technology). The experimental data measured the intensity-intensity time correlation function, which was fitted with Wyatt's proprietary Regularization algorithm that is based on the CONTIN algorithm.[29]

Terahertz spectroscopy

Terahertz time-domain spectroscopy (THz-TDS) measurements were performed using a home-built spectrometer as previously described in detail elsewhere.[30] Briefly, the output of a Ti:Sapphire oscillator (Spectra-Physics Mai Tai SP, 800 nm center wavelength, 84 MHz repetition rate, 3.5 nJ pulse energy, <35 fs pulse duration) was split into two beams for generation and detection of THz radiation using two separate photoconductive antennae (Batop). The time delay between generation and detection beams were controlled using a mechanical delay stage, allowing the transient electric field to be measured.

4 L of bacterial culture was used to produce ∼40 mg of DDX4$^{[1\text{-}245]}$. The solution was centrifuged through the onset of phase separation, yielding tens of microliters of DDX4$^{[1\text{-}245]}$ condensed phase. Samples were then prepared for measurement by pipetting ∼10 μL of condensed phase, dilute phase, or storage buffer solution into the bottom portion of a 100-μm pathlength, two-piece quartz cuvette (Starna Cells). The two-piece cuvettes were sealed with clear nail polish to prevent evaporation of the sample. The sample cuvette was mounted on a holder attached to a mechanical positioning stage, which allowed for the sample measurement, reference measurement through an empty portion of the cuvette, and an air reference measurement to be collected sequentially. The complex refractive index (n) of measurements was processed numerically as previously described[30] and converted to the complex permittivity (ε), or dielectric spectrum, using the relation $\boldsymbol{\varepsilon_S} = \boldsymbol{n}^2$, which holds for nonmagnetic materials.

The resulting permittivity spectra were fit to a constrained double Debye model as shown below in Equation 1, where $\boldsymbol{\varepsilon(\omega)}$ is the frequency-dependent permittivity, $\boldsymbol{\omega}$ is the frequency, $\boldsymbol{\varepsilon_\infty}$ is the dielectric constant in the high-frequency limit, $\boldsymbol{\Delta\varepsilon_i}$ is the relaxation amplitude of the ith Debye resonance ($i = 1, 2$), and $\boldsymbol{\tau_i}$ is the relaxation time of the ith Debye resonance. For the final term corresponding to the intermolecular stretching vibrational mode, $\boldsymbol{A_S}$ is the amplitude, $\boldsymbol{\omega_S}$ is the angular frequency, and $\boldsymbol{\gamma_S}$ is the damping constant.

$$\boldsymbol{\varepsilon(\omega)} = \boldsymbol{\varepsilon_\infty} + \frac{\boldsymbol{\Delta\varepsilon_1}}{\mathbf{1} - \boldsymbol{i\omega\tau_1}} + \frac{\boldsymbol{\Delta\varepsilon_2}}{\mathbf{1} - \boldsymbol{i\omega\tau_2}} + \frac{\boldsymbol{A_S}}{\boldsymbol{\omega_S^2} - \boldsymbol{\omega^2} - \boldsymbol{i\omega\gamma_S}} \quad (1)$$

Because the first Debye resonance ($\boldsymbol{\Delta\varepsilon_1}$) is below the bandwidth of the spectrometer used herein, $\boldsymbol{\tau_1}$ and $\boldsymbol{\tau_2}$ cannot be accurately fit simultaneously. Therefore, the low-frequency Debye resonance lifetime ($\boldsymbol{\tau_1}$) was fixed to 8.76 ps for both H_2O and D_2O based on previous THz-TDS measurements on H_2O in the literature.[31] Likewise, because the intermolecular stretching vibrational mode is higher in frequency than the bandwidth of the spectrometer used herein and has a significant contribution to the total dielectric constant, we fixed $\boldsymbol{\omega_S/2\pi}$ (5.30 THz and 5.36 THz for H_2O and D_2O, respectively) and $\boldsymbol{\gamma_S/2\pi}$ (5.40 THz and 5.06 THz for H_2O and D_2O, respectively) based on literature values.[32] The static dielectric constant ($\boldsymbol{\varepsilon_S}$) was then calculated using the obtained amplitudes from the fit according to Equation 2 as shown below:

$$\boldsymbol{\varepsilon_S} = \boldsymbol{\varepsilon_\infty} + \boldsymbol{\Delta\varepsilon_1} + \boldsymbol{\Delta\varepsilon_2} + \frac{\boldsymbol{A_S}}{\boldsymbol{\omega_S^2}} \quad (2)$$

Stimulated Raman scattering imaging

FPLC-purified DDX4$^{[1\text{-}245]}$ was quenched at room temperature in 100 mM NaCl, 20 mM Tris, 5 mM TCEP (pH = 8). The sample was probed at room temperature using a home-built stimulated Raman scattering (SRS) microscope. SRS uses a laser fixed at the Stokes wavelength (λ_{Stokes}) and a laser tuned to the pump wavelength (λ_{pump}) when the difference $\boldsymbol{\lambda_{pump}^{-1} - \lambda_{Stokes}^{-1}}$ is close to the resonant vibrational frequency.

Two synchronized 6-ps lasers (called pump and Stokes beams) with 80-MHz repetition rate are provided by a picoEmerald system from APE (Applied Physics & Electronics, Inc.). The pump beam is tunable from 720 to 990 nm; the Stokes beam is fixed at 1,064.2 nm. The intensity of the Stokes beam was modulated sinusoidally by a built-in electro-optic modulator at 8 MHz with a modulation depth of more than 90%. Spatially and temporally overlapped pump and Stokes beams were coupled into the laser scanning microscope. After passing through the specimens, forward-going pump and Stokes beams were collected with an IR-coated oil condenser (1.4 NA, Olympus). The Stokes beam is completely filtered with two high-optical-density bandpass filters (890/220 CARS, Chroma Technology), and the transmitted pump beam was detected by a large-area (10 mm × 10 mm) Si photodiode (FDS1010, Thorlabs). The output current of the photodiode was then sent to a fast lock-in amplifier (HF2LI, Zurich Instruments) for signal demodulation. The laser power was set as $P_{\text{pump}} = 17$

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mW, $P_{\text{Stokes}} = 67$ mW; and images were generated with the pixel dwell time of 4 μs, with the time constants of the lock-in amplifier set to 2 μs.

STARLING ensemble generation for DDX4[1-245]

STARLING ensembles were generated using STARLING version 1.0.0 (obtained April 2025) using the VAE/U-Net implementation with DDPM weights model-kernel-epoch.47-epoch_val_loss.0.03.ckpt and VAE weights model-kernel-epoch.99-epoch_val_loss.1.72.ckpt.[33] STARLING is a deep learning generative model trained on data from the Mpipi-GG force field (a derivative of the Mpipi force field) that enables the rapid generation of coarse-grained ensembles of disordered proteins in minutes on commodity hardware.[34] Ensembles contain 6,000 conformers and were generated on a MacBook Pro (M3) using the command: `starling <sequence> -b 400 -c 6000` (note that the `-b` option here defines the batch size, which has no impact on the generated ensemble but scales the number of conformers generated in parallel). Ensemble generation takes ∼15 min per sequence for these large ensembles, although we do not, in general, require ensembles one-tenth of this size for analysis. STARLING ensembles were analyzed using the STARLING package (https://github.com/idptools/starling) and using SOURSOP (http://soursop.readthedocs.io/).[35] Code for generating and analyzing ensembles is available on GitHub at https://github.com/holehouse-lab/supportingdata/tree/master/2025/perets_2025.

All-atom simulations for DDX4[1-245]

All-atom Monte Carlo simulations of the NTD of DDX4 were performed using the CAMPARI simulation engine (version 2; http://campari.sourceforge.net/) in conjunction with the ABSINTH implicit solvent model (parameter set: abs_3.2_opls.prm)[36,37] and the solution ion parameters described by Mao et al.[38] Simulations employed the standard CAMPARI movesets and Hamiltonian parameters, as previously reported.[9,39] Simulations were carried out at 340 K to facilitate robust sampling (as described previously)[9,40] at 5 mM NaCl in a spherical droplet with a fixed radius of 249 Å to minimize finite-size effects. We note these conditions are somewhat distinct from those examined experimentally, and as such, we focus our analysis of CAMPARI simulations primarily on local backbone behavior over long-range interactions. A total of 20 independent simulations were run, each consisting of 30,000,000 equilibration steps followed by 90,000,000 production steps. During the production phase, conformations were saved every 80,000 steps. Across all simulations, this resulted in a final ensemble comprising 23,773 conformers. CAMPARI simulations were analyzed using SOURSOP.[35] CAMPARI simulations took ∼1 month per simulation. Self-avoiding random coil simulations (also referred to as the excluded volume limit) were performed as described previously.[41] Input files for simulations, notebooks for simulation analysis, and trajectory information are all available on GitHub at https://github.com/holehouse-lab/supportingdata/tree/master/2025/perets_2025.

Coarse-grained simulations for DDX4[1-245]

Coarse-grained simulations of DDX4 (Figure S2) were performed using the LAMMPS simulation and the Mpipi-GG force field, a model derived from the Mpipi model.[34,42,43] Initial configurations for all intrinsically disordered proteins were generated as random walks. Prior to production, up to 1,000 iterations of energy minimization were conducted for each system, continuing until the maximum force dropped below 1×10^{-8} (kcal/mol)/Å. Simulations were performed in the NVT ensemble at a NaCl concentration of 150 mM. The temperature was maintained at different values for the coil-to-globule plot using a weakly coupled Langevin thermostat, with adjustments applied every 100 ps. All simulations employed a 20-fs integration timestep. Periodic boundary conditions were used in cubic boxes with a side length of 500 Å. Each simulation was run for 12 milliseconds, with trajectory snapshots saved every 2 μs, generating ensembles of 6,000 conformers per temperature. A single simulation per temperature was run, but the extremely long timescale of these simulations, coupled with long gaps between snapshots, means each conformer is effectively independent of the preceding one, as reflected by the extremely smooth coil-to-globule curve. Input files for simulations, notebooks for simulation analysis, and trajectory information are all available on GitHub at https://github.com/holehouse-lab/supportingdata/tree/master/2025/perets_2025.

Disorder prediction

Disorder prediction (Figure S3) was performed using metapredict (V3).[44]

RESULTS

DDX4[1-245] is intrinsically disordered but exhibits spectroscopic signatures that deviate from those of a random coil

Prior work has established that the DDX4 NTD forms homotypic biomolecular condensates, yielding DDX4 NTD molecules in the dilute and condensed phases (Figure 1b). Using nuclear magnetic resonance (NMR) spectroscopy, Nott et al. found that DDX4[1-236] is an intrinsically disordered protein that remains disordered in the condensed and dilute phases after homotypic phase separation *in vitro*.[26] Moreover, Brady et al. revealed that DDX4[1-236] backbone dynamics in both the condensed and dilute phases undergo fast local reconfiguration.[27] These prior works largely excluded that the DDX4 NTD possesses stable

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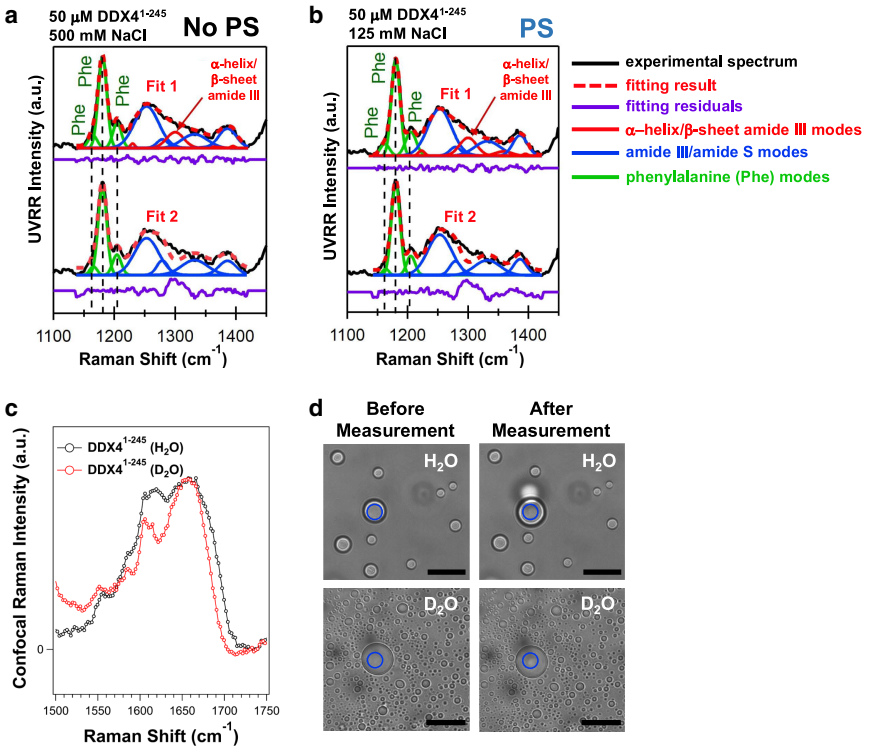


Figure 2. DDX4[1-245] is an intrinsically disordered protein with nonrandom-coil structure. (a and b) UV resonance Raman (UVRR) spectra (λ_{ex} = 215 nm, 5-min total acquisition, 25°C) of (a) one-phase solution of DDX4[1-245] (No PS) or (b) two-phase solution of DDX4[1-245] condensates (PS). Signal from tryptophan was removed via subtraction of the UVRR spectrum of the model compound (L-)Trp. Black curves are experimental spectra, and Gaussian decompositions are shown. The red dashed curves are the sum of individual Gaussian vibrational contributions from phenylalanine (Phe) modes (solid green curves), amide III modes corresponding to α helix/β sheet structure (solid red curves), and additional amide S (1,386 cm^{-1}) and amide III modes (solid blue curves). The two data sets shown in each panel are the same experimental data fit either (top) with or (bottom) without modeling the α helix/β sheet-associated amide III vibrational response. The corresponding residuals from each fit are shown below each spectrum in purple. (c) Confocal spontaneous Raman spectra (λ_{ex} = 532 nm, 25°C) of DDX4[1-245] condensates prepared in (black) H_2O and (red) D_2O measured in the amide I vibrational region. Each spectrum is an average of 10 acquisitions, with each acquisition in H_2O or D_2O collected for an exposure time of 55 s or 40 s, respectively. (d) Bright-field images of DDX4[1-245] condensates, with regions measured in (c) marked by the blue circles. Samples are shown before and after laser exposure (~10 min). Scale bars, 20 μm.

secondary or tertiary structure in the condensed or dilute phases. However, it is not known whether the DDX4 NTD behaves as a self-avoiding random coil (i.e., a chain with excluded volume but no meaningful attractive or repulsive interactions)[45] or possesses sequence-encoded local and long-range interactions.

To investigate the conformational properties of DDX4[1-245] in the dilute and condensed phases, we applied UV resonance Raman spectroscopy and confocal Raman microscopy (Figures 2a–2c). In UV resonance Raman spectroscopy, the excitation wavelength is resonant with electronic transitions in the protein, making this technique a sensitive probe of aromatic protein side chains and the amide backbone. In particular, the amide III mode, which is the coupled C-N stretching and N-H bending along the protein backbone, can be used to determine protein secondary structures.

The UV resonance Raman spectra of single-phase solutions (Figure 2a, black curves) and phase-separated solutions (Figure 2b, black curves) of purified, unlabeled DDX4[1-245] *in vitro* are similar (see Table S1 for fitting parameters of the spectra). This suggests that DDX4[1-245] does not undergo significant changes in secondary structure upon phase separation. Both single-phase and phase-separated solutions of DDX4[1-245] show strong UV resonance Raman bands at 1,160, 1,180, and 1,205 cm^{-1} from phenylalanine (Figures 2a and 2b, green curves). The amide III and amide S (C_α-H bending) modes that are sensitive to protein secondary structure appear between 1,250 and 1,350 cm^{-1} and at 1,386 cm^{-1}, respectively (Figures 2a and 2b, blue curves). The amide III mode at 1,253 cm^{-1} is characteristic of conformations commonly seen in β sheets, whereas the amide S mode at 1,386 cm^{-1} can be present in both β sheet and random coil configurations.[46] An amide III vibrational mode at 1,300 cm^{-1} is only expected for α helix or β sheet secondary structures, but not for random coil conformational behavior. The residuals (Figures 2a and 2b, purple curves) comparing alternative fits 1 and 2 (Figures 2a and 2b, dashed red curves) to the experimental data (Figures 2a and 2b, black curves) show that the experimental spectra of single-phase and two-phase solutions have inferior fits in the absence of a vibrational mode at 1,300 cm^{-1} (Figures 2a and 2b, bottom, dashed red curves). The presence of an amide III vibrational band at 1,300 cm^{-1} (Figures 2a and 2b, top, solid red curves) suggests DDX4[1-245] deviates from total random coil conformational behavior, expanding on prior NMR studies.[26,27]

To probe the Raman response inside individual unlabeled DDX4[1-245] condensates, we used confocal Raman microscopy (Figure 2c) with a visible laser ($\lambda_{excitation}$ = 532 nm) and bright-field imaging (Figure 2d). The amide I vibrational mode, primarily C=O stretching of the protein backbone, of phase-separated DDX4[1-245] condensates

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peaks at 1,660 cm^{-1} (Figure 2c, black curve). The broad amide I vibrational response spanning 1,600–1,700 cm^{-1} could suggest contributions from α helix and β sheet secondary structures as well as random coil conformational behavior.[47] However, the bending signal from H_2O inside condensates can couple with the amide I vibrational response.[48] Accurate assignment of the amide I Raman response thus requires deconvolution of spectral contributions from protein and the bending mode of water within condensates. Preparation of DDX4$^{[1-245]}$ condensates in D_2O shifts spectral contributions of water bending out of the amide I region (Figure 2c, red curve), resolving the Raman contributions of arginine and aromatic tyrosine and phenylalanine side chains ∼1,550–1,610 cm^{-1} and the amide I response that spans ∼1,600–1,690 cm^{-1} and peaks at 1,660 cm^{-1}. Nonetheless, we cannot exclude the possibility that the spectral change at ∼1,670 cm^{-1} is due to DDX4$^{[1-245]}$ exhibiting a unique secondary structure in D_2O. These observations are consistent with secondary structure conformational behavior that deviates from exclusively random coil conformations. Consistent with these spectroscopic results, all-atom simulations identified little persistent secondary structure but found local deviations from self-avoiding random coil conformational behavior (Figures S3–S5). Overall, our results using multiple approaches of UV resonance Raman, confocal Raman microscopy, and computation each suggest that DDX4$^{[1-245]}$ is disordered yet possesses local conformational biases that deviate its backbone behaviors from those of a random coil.

Having investigated local conformational behavior using spectroscopic and computational methods, we next asked if the conformational behavior of monomeric DDX4$^{[1-245]}$ under dilute conditions (i.e., the DDX4$^{[1-245]}$ ensemble) possessed interactions that could cause the observed protein structural deviations away from a self-avoiding random coil. To examine this question, we turned to generating a computational molecular model of ensembles using STARLING, a deep-learning method for predicting conformational ensembles of intrinsically disordered regions.[33] STARLING-derived ensembles predict a hydrodynamic radius of 41.3 Å (Figure S5), in good agreement with all-atom simulations (Figure S4), and reveal substantial intramolecular interactions (Figure S6). These results suggest that while DDX4$^{[1-245]}$ is disordered, sequence-encoded interactions lead to deviations from self-avoiding random coil-like behavior.

To investigate the molecular interactions in the DDX4$^{[1-245]}$ ensemble, we used STARLING to generate ensembles for previously characterized phase-separation-deficient mutants: an all arginine-to-lysine variant (24RtoK, disrupting native cation-π interactions) and an all phenylalanine-to-alanine variant (14FtoA, disrupting cation-π and π-π interactions). Both ensembles reveal a substantial reduction of interactions (Figure S6), albeit with distinct impacts on the ensemble. They also showed a corresponding increase in hydrodynamic radius (R_h) (Figure S7). We then generated an ensemble for a previously uncharacterized mutant that replaces all negatively charged residues with alanine (37EDtoA, disrupting electrostatic interactions), which again yielded an ensemble with a reduction of interactions and an increase in R_h compared with the wild-type sequence (Figures S6 and S7). Taken together, we interpret these results to mean that intramolecular interactions in DDX4$^{[1-245]}$ are driven by multiple distinct modes of interaction, including the π-π, cation-π, and electrostatic interactions.

DDX4$^{[1-245]}$ monomer collapse correlates with phase separation

We next employed DLS to characterize monomeric DDX4$^{[1-245]}$ under both non-phase-separating and phase-separating conditions. Specifically, we sought to understand how the global dimensions of DDX4$^{[1-245]}$ change as a function of temperature, protein concentration, and ionic strength (Figure 3) when the system crosses the phase boundary. Nott et al.,[26] Brady et al.,[27] and Lin et al.[28] demonstrated that the DDX4 NTD undergoes homotypic phase separation to form condensates that are quantitatively described by physical models for phase separation. Moreover, this prior work established that homotypic phase separation of DDX4 NTD is suppressed at higher salt concentrations and temperatures.[25,26] We therefore sought to understand how phase separation, as promoted or repressed by changes in parameters of temperature, protein concentration, and ionic strength, altered the conformational behavior of DDX4$^{[1-245]}$.

At a single-phase condition (100 μM DDX4$^{[1-245]}$, 200 mM NaCl, 25°C), DLS reveals a single population of DDX4$^{[1-245]}$ with apparent mean radius of hydration $\langle R_h \rangle = 4.2 \pm 1.1$ nm (Figures 3a and 3b, top). This value is in good agreement with STARLING-derived ensembles with $\langle R_h \rangle = 4.1$ nm (Figure S8). Our experimentally measured $\langle R_h \rangle = 4.2 \pm 1.1$ nm is also consistent with the reported value for the DDX4 N terminus of 3.16 nm measured using pulsed-field gradient NMR.[27] The dispersity of the distribution ±1.1 nm is given by fitting the DLS intensity-intensity correlation function. Given the large dispersity relative to the measured size of 4.2 nm, we cannot rule out that DDX4$^{[1-245]}$ oligomers in the dilute phase also contribute to the distribution.[49] Under conditions in which phase separation is suppressed and protein concentration is increased (275 μM DDX4$^{[1-245]}$, 300 mM NaCl, 10°C), the $\langle R_h \rangle$ grows to 5.4 ± 1.3 nm. One possible interpretation of this result is that higher protein concentrations may favor the formation of DDX4$^{[1-245]}$ oligomers (Figure 3c, top) (see also discussion).

Next, we sought to understand what happens to DDX4$^{[1-245]}$ monomer under phase-separating conditions. Phase separation of DDX4$^{[1-245]}$ was induced by cooling the samples to 10°C (Figures 3a–3c, center). Surprisingly, under conditions

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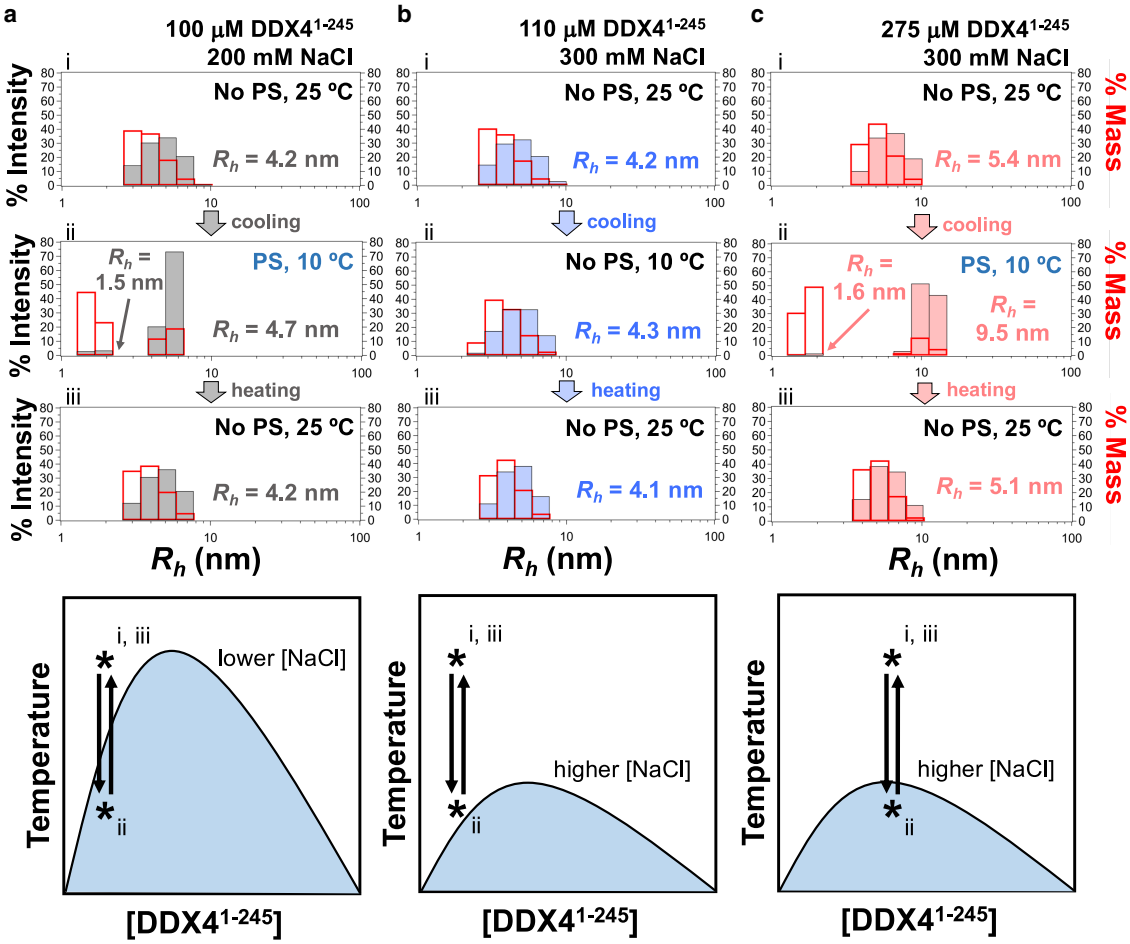


Figure 3. DDX4$^{[1-245]}$ chain collapse correlates with phase separation. (a–c) Representative dynamic light scattering measurements of monomeric DDX4$^{[1-245]}$ as function of temperature, protein concentration, and NaCl concentration: percent light scattering intensity of DDX4$^{[1-245]}$ (filled gray, blue, and pink bars) is on the left axis, and percent mass of DDX4$^{[1-245]}$ is on the right axis (unfilled red bars). Mean radius of hydration (R_h) is reported for each observed population. Lower panels show schematic DDX4$^{[1-245]}$ binodals (based on Brady et al.[27]) and corresponding experimental cycles of (i) starting point before cooling, (ii) stopping point after cooling, and (iii) endpoint after heating, with each point marked by an asterisk.

at 10°C where DDX4$^{[1-245]}$ crossed the phase diagram binodal (Figures 3a and 3c),[26,27] we observed that ∼70%–80% formed a distinct protein population with dramatically reduced apparent $\langle R_h \rangle = 1.6 \pm 0.2$ nm. We interpret this result to reflect compaction of DDX4$^{[1-245]}$ monomer chains under phase-separating conditions. This DDX4$^{[1-245]}$ collapsed state is surprisingly small for a 245-residue intrinsically disordered protein (Figure S9). However, the globular protein mCherry of comparable length and molecular weight (236 residues, 26.7 kDa) was reported by Pappu and Brangwynne groups to have $R_h = 1.4$ nm.[50] In addition, a scaling law proposed by Schuler and co-workers for densely compacted proteins ($\boldsymbol{R_h} = \frac{\rho_0 N^{1/3}}{0.77}$, where N is the number of bonds, and the empirically determined scaling factor $\rho_0 = 0.2$) predicts $R_h = 1.6$ nm for DDX4$^{[1-245]}$.[51] Finally, in our coil-to-globule MD simulations for DDX4$^{[1-245]}$ the most compact coarse-grained ensembles give R_h of 2.2 nm (Figure S2), which is near to our experimentally measured value, supporting a highly compact state of DDX4$^{[1-245]}$.

In principle, coil-to-globule chain collapse is expected at lower temperatures for polymers with an upper critical solution temperature (i.e., enthalpy-driven chain collapse)[52,53] such as DDX4$^{[1-245]}$ (Figure S2).[26] We note that monomer collapse was not observed where DDX4$^{[1-245]}$ is cooled under higher salt concentrations and does not cross the binodal (Figure 3b, center). Importantly, the collapsed monomer population also predominates under phase-separating conditions, accounting for an estimated 70%–80% of the sample (percent mass reported in Figure 3 with red bars). Cycling the temperature to 25°C and re-crossing the binodal shows that entry and exit of the monomer from this collapsed state were also reversible, a hallmark of phase separation. Thus, across conditions of temperature, protein concentration, ionic strength, and without the use of any molecular labels, we conclude from DLS that the appearance of a new, highly compact DDX4$^{[1-245]}$ monomer population correlates strongly with DDX4$^{[1-245]}$ phase separation.

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Dielectric constant is reduced inside DDX4[1-245] condensates

Confocal Raman microscopy showed no significant structural changes of phase-separated DDX4[1-245] in H_2O or D_2O (Figure 2c). However, under equivalent solution conditions, DDX4[1-245] in D_2O phase separates at a lower protein concentration than in H_2O (Figure 4a). In 500 mM NaCl at 25°C, one order of magnitude higher protein concentration is needed for DDX4[1-245] phase separation in H_2O versus D_2O. As DDX4[1-245] exhibits no secondary structure changes in D_2O versus H_2O (Figure 2c), our observations suggest that water plays a key role in tuning the phase boundaries of DDX4[1-245] (see also discussion).[27,54]

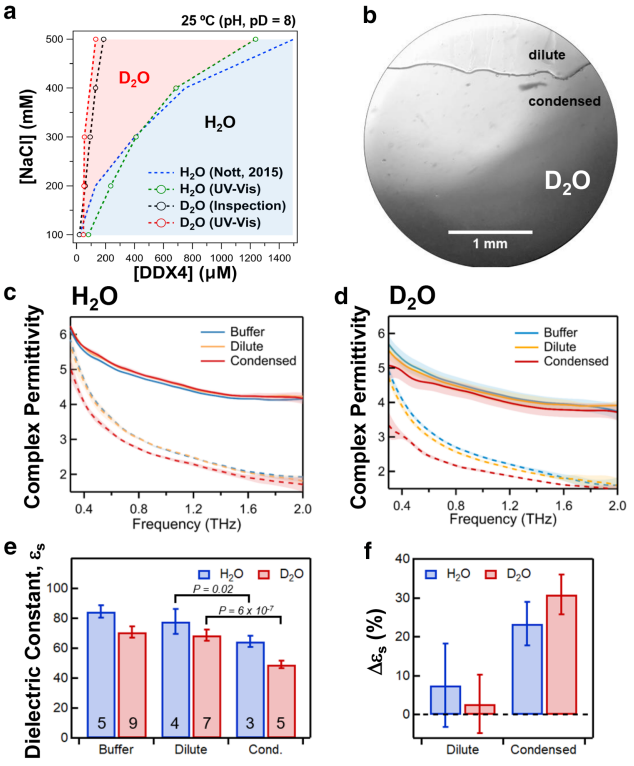


Figure 4. Terahertz-based quantification of the complex permittivity of monomeric and condensed DDX4[1-245]. (a) Phase diagram binodal of DDX4[1-245] measured in (blue) H_2O or (red) D_2O, constructed based on Nott et al.[26] (blue dashed curve), UV-visible absorbance of the dilute phase (green and red dashed curves, "UV-Vis"), or bright-field microscopy (black dashed curve, "Inspection"). (b) Spontaneous separation of DDX4[1-245] in D_2O into (top) dilute and (bottom) condensed phases within the terahertz (THz) cuvette. (c and d) Complex permittivity of buffer (blue), dilute (yellow), and condensed phases (red) of phase-separated DDX4[1-245] prepared in H_2O or D_2O measured by terahertz (THz) time-domain spectroscopy. The dashed and solid lines represent the imaginary and real components of the complex permittivity, respectively. The shaded area represents the standard deviation across multiple measurements. (e) From (c) and (d), the static dielectric constant (ε_s) of buffer, dilute, and condensed phases in H_2O (blue) or D_2O (red). The number of measurements collected for each condition is also shown. (f) The percent difference of $\boldsymbol{\varepsilon_S}$ between dilute and condensed phases in H_2O or D_2O compared with buffer.

Using NMR, Brady et al. previously estimated that the static dielectric constant (ε_s) of the *in vitro* DDX4[1-236] condensed phase could range from 20 to 47 (as compared with 80 for pure H_2O).[27] However, the complex permittivity (also known as the dielectric response) inside condensates has never been directly quantified. Therefore, we aimed to directly measure the complex permittivity inside DDX4[1-245] condensed phase using terahertz (THz) spectroscopy.

THz time-domain spectroscopy (THz-TDS) measures the phase-resolved transmission of coherent far-infrared radiation (~1 THz = 33 cm^{-1}), enabling the direct extraction of the frequency-dependent complex refractive index, and therefore permittivity, without requiring a Kramers-Kronig approximation (Figure 1d). In aqueous solutions, the dielectric loss in the THz region is dominated by Debye relaxation of water molecules aligned by the transient electric field. The phase-resolved transmission of THz radiation by water is used to extract the frequency-dependent complex refractive index (n) of the solution, which is equal to the square root of the complex permittivity (dielectric response, $\boldsymbol{\varepsilon}$) (Figure 1d).[30] The resulting dielectric spectrum can then be fit to a constrained Debye model to semi-quantitatively extract the static permittivity (i.e., dielectric constant, $\boldsymbol{\varepsilon_S}$). We concentrated DDX4[1-245] via centrifugation in H_2O and D_2O to induce phase separation, then carefully transferred either the dilute or condensed layer to a quartz microcuvette by pipetting (Figure 4b), and measured the complex-valued permittivity spectra (0.25–2.0 THz) of the dilute phase, the condensed phase, and the buffer for both H_2O and D_2O (Figures 4c and 4d).

In both H_2O and D_2O, the imaginary component of the complex permittivity (Figures 4c and 4d, dashed lines) was suppressed in the DDX4[1-245] dense phase. Fitting the data to a constrained double Debye model (see materials and methods and Tables S2 and S3) gave static dielectric constants of pure buffer in the expected range ($\boldsymbol{\varepsilon_s^{buffer,\,H_2O}} = 84 \pm 4$ and $\boldsymbol{\varepsilon_s^{buffer,\,D_2O}} = 71 \pm 4$), whereas the condensed phase static dielectric constant ($\boldsymbol{\varepsilon_s^{condensed,\,H_2O}} = 67 \pm 6$ and $\boldsymbol{\varepsilon_s^{condensed,\,D_2O}} = 49 \pm 2$) was significantly reduced relative to the dilute phase (Figure 4e). For buffer solutions, our static dielectric constant is overestimated by ~7% for H_2O based on tabulated values ($\boldsymbol{\varepsilon_s^{lit,\,H_2O}} = 78.36$ at 25°C) but is underestimated by ~9% for D_2O ($\boldsymbol{\varepsilon_s^{lit,\,D_2O}} = 77.936$ at 25°C).[55] Because THz-TDS measures the "high-frequency" dielectric response, measurement of the static dielectric constant relies on changes in the amplitudes and lifetimes of the Debye relaxation terms used in the fitting model. Relative to buffer, the percent change of static dielectric in the condensed phase corresponded to a change of ~20% in H_2O and ~30% in D_2O (Figure 4f), indicating that our measurement is still sensitive to the relative change in static dielectric constant upon condensation at a semi-quantitative level. Overall, our analysis suggests that although $\boldsymbol{\varepsilon_s^{condensed,\,H_2O}}$ is substantially lower than in the dilute phase, the reduction of $\boldsymbol{\varepsilon_s^{condensed,\,H_2O}}$ is less significant than previously predicted.[26,27] In sum, relative to the surrounding

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bulk aqueous environment, condensates offer unique chemical environments. Moreover, the solvent itself can have a major impact on the intracondensate chemical environment, as seen here for H_2O versus D_2O.

Optical observation of reduced water content inside DDX4[1-245] condensates

Prior NMR experiments and surface wetting measurements suggested ~75% volume fraction of water inside DDX4[1-236] condensates.[27,56,57] We aimed to directly image the reduction of water content inside DDX4[1-245] condensates using SRS imaging (Figure 5). Whereas conventional spontaneous Raman scattering emits relatively few inelastically scattered photons, SRS can achieve an enhancement factor as high as $\sim 10^8$ over spontaneous Raman.[58] Our SRS imaging achieves point scanning with a lateral resolution of ~330 nm and an axial resolution of ~1.4 μm.

We imaged DDX4[1-245] condensates in H_2O using SRS of the CH_3 stretching (Figure 5a) and the O-H stretching (Figure 5b) vibrational modes. The O-H stretching vibration showed a strong SRS response both inside and outside the condensates because the overwhelming contribution was due to water (Figure 5b). Compared with the dilute phase, DDX4[1-245] condensates appear darker in SRS images, clearly showing a reduction of water vibrational response (Figure 5b). The interface of condensates appears even darker compared with the condensate core (Figure 5c). The measured SRS intensity of water O-H stretching within the condensate core was ~2.5% lower compared with bulk, whereas at condensate perimeters, it was ~4% lower (Figure 5d). NMR and surface wetting measurements previously suggested the water content in the DDX4[1-245] condensed phase is reduced by 25% compared with the dilute phase,[27,56,57] but the reduction of the SRS water O-H stretching vibrational response is not as large. Nevertheless, the changes in the intensity of the water vibrational signal in the SRS image, as well as our THz measurements that registered an ~20% reduction of the dielectric, support a substantial reduction in water content inside DDX4[1-245] condensates.

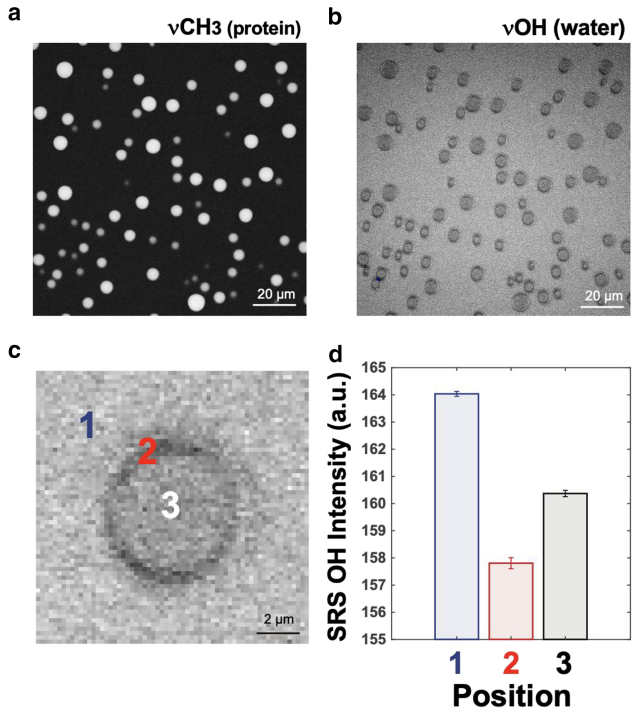


Figure 5. Stimulated Raman scattering (SRS) of reduced water response in DDX4[1-245] condensates. (a) SRS imaging (λ_{Stokes} = 1,064.2 nm, λ_{pump} = 810.5 nm) detects CH_3 vibrational response (2,941 cm^{-1}) of DDX4[1-245] protein concentrated in condensates. (b) SRS imaging (λ_{Stokes} = 1,064.2 nm, λ_{pump} = 790 nm) reveals reduced O-H vibrational response (3,261 cm^{-1}) of water inside and at the interface of DDX4[1-245] condensates. (c) Detail of an individual condensate O-H vibrational response imaged with SRS. (d) Quantification (mean ± SEM) of SRS water O-H intensity at positions (1) outside, (2) at the interface, and (3) inside condensates for $N = 67$ condensates.

Summary of results

Label-free spectroscopy of condensates together with deep-learning based computational studies of protein dynamics and interactions have yielded new observations of DDX4[1-245] phase separation. Our results suggest DDX4[1-245] is disordered in the dilute and condensed phases but deviates from random coil behavior due to sequence-encoded intramolecular interactions. We observed DDX4[1-245] protein chain collapse upon crossing the phase diagram binodal. Inside DDX4[1-245] condensates, the complex permittivity (dielectric) is significantly reduced. We demonstrated that this reduction of the dielectric is even more pronounced when condensates are prepared in D_2O, and that the DDX4[1-245] phase diagram is significantly shifted in D_2O versus H_2O. Finally, we imaged how DDX4[1-245] condensates exclude water, a process likely to strengthen protein-protein interactions.

DISCUSSION

We applied UV resonance Raman, confocal Raman, SRS, and terahertz (THz) spectroscopies in combination with DLS, as well as computational ensemble prediction methods to probe the conformational behavior of individual proteins and the chemical and physical properties of condensate interiors and interfaces. Our application of this suite of complementary, quantitative methods leads us to propose a molecular model of DDX4[1-245] phase separation (Figure 6). Our model comprises protein monomer collapse, water release, and reduction of the dielectric. Further kinetic studies will be important to validate the sequence of these molecular events.

For one-phase solution conditions, DDX4[1-245] exists as monomers that may form small oligomers at high protein concentration. Using DLS, we detect a single broad

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population with apparent mean radius of hydration $\langle R_h \rangle$ = 4.2 ± 1.1 nm (Figures 3a and 3b), which increases with increasing protein concentration (Figure 3c). This hydrodynamic radius is in good agreement with ensembles generated using STARLING ($\langle R_h \rangle$ = 4.1 nm) (Figure S8). STARLING-derived ensembles for DDX4$^{[1\text{-}245]}$ suggest attractive interactions driven by cation-π, π-π, and electrostatic interactions. Such interactions cause deviations in the single-chain ensemble away from the behavior expected for a real chain with excluded volume but no meaningful attractive or repulsive interactions (i.e., a self-avoiding random coil). Moreover, DDX4 variants deficient in phase separation show a substantial diminution in intramolecular interactions, consistent with prior work.[9,59,60]

The observation of chain deviations from self-avoiding random coil conformational behavior is consistent with our analysis of UV resonance Raman amide III response (Figures 2a and 2b). Prior NMR work on the DDX4 NTD excluded the possibility of the stable long-lived acquisition of secondary structure.[27] Nevertheless, our UV resonance Raman amide III response is incompatible with a conformational ensemble that is exclusively random coil, suggesting that nonrandom intramolecular interactions give rise to conformations that deviate from random coil configurations. In addition, UV resonance Raman probing the amide III response (Figures 2a and 2b) and confocal Raman microscopy probing the amide I response (Figure 2c) both suggest that DDX4$^{[1\text{-}245]}$ molecules retain their dilute-phase conformational biases in the condensed phase.

Under conditions in which phase separation can occur for DDX4$^{[1\text{-}245]}$ (e.g., reduced temperature and reduced salt concentration), our DLS measurements show that DDX4$^{[1\text{-}245]}$ condensates grow according to Ostwald ripening dynamics (Figure S10). Our model proposes that DDX4$^{[1\text{-}245]}$ monomers can also undergo collapse ($\langle R_h \rangle$ = 1.6 ± 0.2 nm) under phase separation conditions (Figure 3; Figures S2 and S11). For polymers with upper critical solution temperature (UCST), such as DDX4$^{[1\text{-}245]}$, phase separation is expected to be enthalpy driven, and chain collapse is expected.[26,52,53] Most reported biological condensates have an UCST, so our model of chain collapse and water release could potentially represent a general mechanism of protein phase separation.

In addition to monomeric DDX4$^{[1\text{-}245]}$, we also observe the appearance of larger species by DLS ($\langle R_h \rangle$ = 5.4 nm ± 1.3 nm) even under conditions in which phase separation is suppressed (Figure 3c). One possible explanation for these larger species is that they are small oligomers (estimated to be dimers or trimers) of DDX4$^{[1\text{-}245]}$ formed from monomeric DDX4$^{[1\text{-}245]}$ that retain dimensions of around 4.2 ± 1.1 nm. An alternative interpretation is that these larger species are clusters of tens of highly compact DDX4$^{[1\text{-}245]}$ monomers, where each monomer in the cluster has an R_h closer to the compact population observed under phase-separating conditions. Such an interpretation could be consistent with prior work in which clusters of disordered proteins were observed under subsaturated conditions.[49] However, even in that prior work, a small increase in the dimensions of monomers was interpreted as small oligomers, whereas clusters were found to be an order of magnitude larger than monomers. As such, these findings currently favor the simpler explanation of small oligomers. Methods such as label-free high-resolution mass photometry[61] could possibly resolve these possibilities.

The DDX4$^{[1\text{-}245]}$ collapsed monomer can be considered a single-chain condensate, in other words, the smallest possible condensate unit. The parallels between single-chain condensation and multi-chain phase separation have been drawn since the 1970s.[62–64] For synthetic polymers, single-chain collapse across conditions that mirror the phase boundary was also reported.[65] Recently, time-resolved SAXS revealed single-chain compaction of the low-complexity prion-like domain from the RNA-binding protein hnRNPA1 upon transition into phase-separation-compatible conditions and prior to coalescence and growth.[66] This work focused on the kinetics of phase

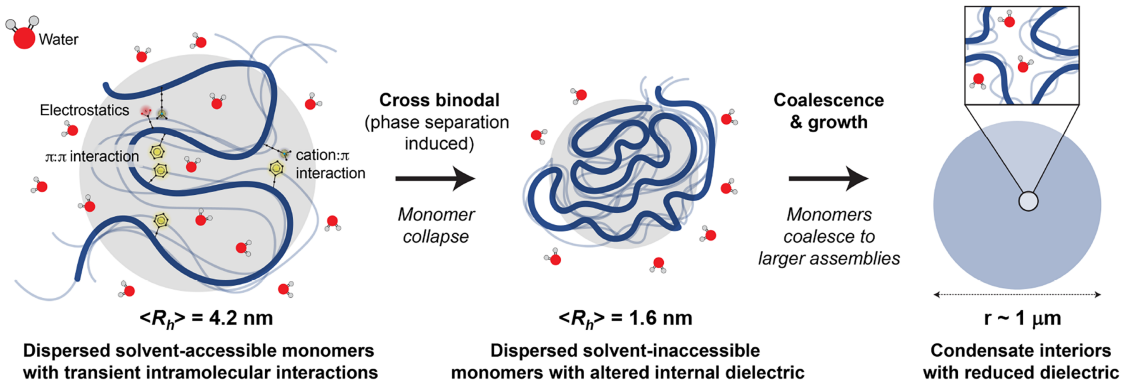


Figure 6. Model of DDX4$^{[1\text{-}245]}$ liquid-liquid phase separation. Our results from terahertz spectroscopy, stimulated Raman scattering imaging, and dynamic light scattering suggest that phase separation (PS) triggers DDX4$^{[1\text{-}245]}$ protein chain collapse, water release, and reduction of the static dielectric constant ($\boldsymbol{\varepsilon_S}$). A reduced dielectric and water release can increase the strength of protein-protein cation-π and π-π interactions inside condensates, driving condensate growth and phase separation, possibly via positive feedback.

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separation and suggested this compact species was relatively transient. In contrast, here, we observe a population of compact monomers that persists for at least timescales accessible to DLS without the need for a stopped-flow or microfluidics apparatus. As such, we propose that the DDX4$^{[1\text{-}245]}$ collapsed state may represent an experimentally accessible target for studying early-stage condensate nucleation and growth. Further characterization and validation of the DDX4$^{[1\text{-}245]}$ collapsed state, potentially via NMR or single-molecule fluorescence resonance energy transfer measurements, may offer a powerful model system to dissect the specific molecular interactions that underlie the kinetics, thermodynamics, and chemical driving forces of condensate nucleation and growth.

As reported previously, DDX4$^{[1\text{-}245]}$ is overrepresented in phenylalanine and arginine residues compared with other intrinsically disordered proteins, and cation-π and π-π interactions appear to be critical to the protein's phase behavior.[1,11,25–27] STARLING ensembles support the importance of these interactions and show that losing either native arginine residues (24RtoK mutant) or native phenylalanine residues (14FtoA mutant) leads to an increase in global dimensions and loss of intramolecular interactions (Figures S6 and S7). Moreover, our work predicts that the loss of acidic residues will also suppress intramolecular interactions, reducing homotypic phase separation (Figures S6 and S7). The importance of arginine-aromatic interactions in driving condensate formation for intrinsically disordered regions is well established, with arginine-to-lysine substitutions shown in many contexts to abrogate attractive interactions essential for phase separation.[8,26,27,67–70] Prior work suggests that arginine-phenylalanine interactions are stronger than lysine-phenylalanine interactions.[71] Computational work has indicated that a major determinant of arginine's importance (compared with lysine) stems from the difference in free energy of hydration between the residues.[72] The calculated free energy of hydration difference (∼12 kcal/mol) between positively charged arginine and lysine residues shows that arginine is more easily dehydrated, consistent with a model in which protein dehydration and water release from arginine-rich condensates can help to promote phase separation. Therefore, besides protein chain collapse, any proposed model must also account for how water behavior may drive DDX4$^{[1\text{-}245]}$ phase separation.

Our development of THz-TDS and *in situ* SRS imaging of condensed phase DDX4$^{[1\text{-}245]}$ (Figures 4 and 5) has enabled us to directly probe water release and reduced static dielectric, which can drive phase separation and tune protein-protein interactions inside condensates. Inside the DDX4$^{[1\text{-}245]}$ condensed phase, the dielectric is reduced to $\boldsymbol{\varepsilon_s^{condensed,\,H_2O}} \leq 67 \pm 6$ and $\boldsymbol{\varepsilon_s^{condensed,\,D_2O}} \leq 49 \pm 2$ (Figure 4e), and SRS imaging shows reduced intensity of water vibrational signals (Figure 5d). The reduction of the dielectric can be explained if the dielectric response of water in close proximity to DDX4$^{[1\text{-}245]}$ differs markedly from the dielectric response of pure water. The high dielectric value of free water is related to the free rotation of the static dipoles of water molecules in an external field. Our observation of reduced dielectric response of water inside condensates strongly suggests that these water molecules are not freely rotating and form either a solvation shell or a caged geometry, consistent with recent reports from 2D-IR spectroscopy of water dynamics inside condensates.[73] In both cases, the polarizability per molecule will be lower, resulting in the reduced overall dielectric response. Our results and interpretations are consistent with recent measurements of condensate dielectrics using fluorescent dyes, which suggested that the observed reduction of the dielectric and the inferred lower hydrophobicity across multiple protein condensate systems could not be explained by lower water content.[74] Interestingly, the same work showed that the dielectric of condensates governs interactions with membrane surfaces, suggesting another functional role for the dielectric in condensate biology.

Cation-π and π-π protein-protein interactions are believed to help drive DDX4$^{[1\text{-}245]}$ phase separation. In the aqueous phase dominated by the high dielectric constant of H_2O (∼80), it has been shown that the cation-π interaction energy is similar to a single hydrogen bond (∼5 kcal/mol), but it becomes favorable (∼19 kcal/mol in the gas phase) as the dielectric of the medium is reduced.[75,76] Computational predictions suggest that our experimentally observed changes in dielectric inside DDX4$^{[1\text{-}245]}$ condensates ($\boldsymbol{\varepsilon_s^{condensed,\,H_2O}} \leq 67 \pm 6$ and $\boldsymbol{\varepsilon_s^{condensed,\,D_2O}} \leq 49 \pm 2$) would increase the interaction strength of individual cation-π interactions by < 0.1 kcal/mol,[76] which emphasizes the importance of multivalent interactions in driving biological phase separation. As condensates form and grow, increasingly stronger cation-π and π-π protein-protein interactions may promote further water release and reduction of the dielectric, likely establishing positive feedback that can drive phase separation and condensate growth. Significantly, our experimental data suggest a model that unifies the concepts of protein chain collapse, reduced dielectric, and protein dehydration as mechanisms that strengthen multivalent protein-protein interactions to drive phase separation.

Our SRS imaging measurements also revealed reduced O-H stretching signal from water at the condensate interface compared with the condensate core (Figures 5c and 5d). Notably, SRS imaging of CH_3 vibrations did not resolve any difference in protein content between the core and the interface of DDX4$^{[1\text{-}245]}$ condensates (Figure 5a). This result implies that the reduction in O-H stretching signal from water at the condensate interface does not result from increased protein content or lensing artifacts in our SRS imaging experiments. This observation might suggest either a lower water content or changes in the chemical environments of interfacial water that lead to spectral changes in the O-H stretches, consistent with recent reports demonstrating that strong electric fields can emerge at the interfaces of condensates and oil-water emulsions and give rise

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to distinct organization of molecules at interfaces.[14,77–79] Whether our findings about interfacial water can be generalized to other biological condensate systems requires further study.

Surprisingly, we also observed that D_2O dramatically shifts the DDX4[1-245] binodal compared with H_2O (Figure 4a). The impact of D_2O on phase separation has been reported previously in the context of a dehydrogenase,[80] lysozyme,[81] aprotinin,[82] γB-crystallin,[83] bovine serum albumin,[84] and androgen receptor proteins,[85] where qualitatively similar observations were reported, but the origin of this isotope effect remains unclear. Neutron scattering studies of protein solubility and phase separation[80–84] observed that attractive protein-protein interactions are strengthened in D_2O, although the physical mechanisms (e.g., stronger hydrophobic or electrostatic interactions) underlying this phenomenon remain debated. Our observation that $\boldsymbol{\varepsilon_s^{condensed,\,D_2O} < \varepsilon_s^{condensed,\,H_2O}}$ (Figure 4) likewise implies stronger attractive protein-protein interactions. Besides protein-protein interactions, D_2O hydrogen bonds are also stronger by 0.1–0.2 kcal/mol compared with H_2O due to differences in their zero-point energies.[86] This energy difference can alter the energy landscape and fine tune the protein concentration dependence of biological phase separation. Both stronger protein-water interactions involving hydrogen-bonding interactions with D_2O and stronger protein-protein interactions will enthalpically ($\Delta\boldsymbol{H^{protein\text{-}water}}$ and $\Delta\boldsymbol{H^{protein\text{-}protein}}$) favor phase separation, counterbalancing the free energy costs of reduced numbers of hydration water molecules and protein configurational entropy ($\Delta\boldsymbol{S^{protein\text{-}water}}$ and $\Delta\boldsymbol{S^{protein\text{-}protein}}$). These conditions are consistent with DDX4[1-245] phase separation being enthalpy driven and suggest that D_2O may also promote DDX4[1-245] chain collapse to drive phase separation.

CONCLUSION

This work demonstrates how label-free vibrational spectroscopies and optical scattering methods provide insights into the underlying physical chemistry of condensate formation. Through integration of light-scattering and computational methods, we identify a collapsed monomeric species that correlates with phase separation. Our measurements quantify critical physical, chemical, and structural properties of phase-separated condensates that until now have proven experimentally refractory and computationally inaccessible. Our data lead us toward a model of phase separation that unifies the molecular concepts of protein dehydration, water release, reduced water polarizability, protein chain collapse, and protein-protein interactions as critical driving forces of homotypic protein phase separation. In particular, the reduction in dielectric and release of water upon condensate formation might strengthen the multivalent protein-protein interactions that drive condensation. These findings lead us to propose a positive-feedback loop that can drive liquid-liquid phase separation of proteins, which could potentially be a common mechanism for protein phase separation.

DATA AVAILABILITY

All data will be shared upon reasonable request. Code for generating and analyzing ensembles, input files for simulations, notebooks for simulation analysis, and trajectory information are all available on GitHub at https://github.com/holehouse-lab/supportingdata/tree/master/2025/perets_2025.

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AUTHOR CONTRIBUTIONS

E.A.P. designed research, performed research, analyzed data, supervised research, and wrote the manuscript; J.A.S. wrote the manuscript, performed research, and analyzed data; J.H.C. performed research and analyzed data; L.S. performed research and analyzed data; D.K.A. performed research and analyzed data; A.S.H. performed research, analyzed data, contributed analytic tools, and wrote the manuscript; J.E.K. analyzed data and supervised research; W.M. supervised research; J.N. performed research and supervised research; E.C.Y.Y. designed research, supervised research, and wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPLEMENTAL INFORMATION

Supporting material can be found online at https://doi.org/10.1016/j.bpj.2026.07.018.

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Supplemental information

Chain collapse, reduced dielectric, and water release drive protein phase separation

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Supporting Information

Chain Collapse, Reduced Dielectric, and Water Release Drive Protein Phase Separation

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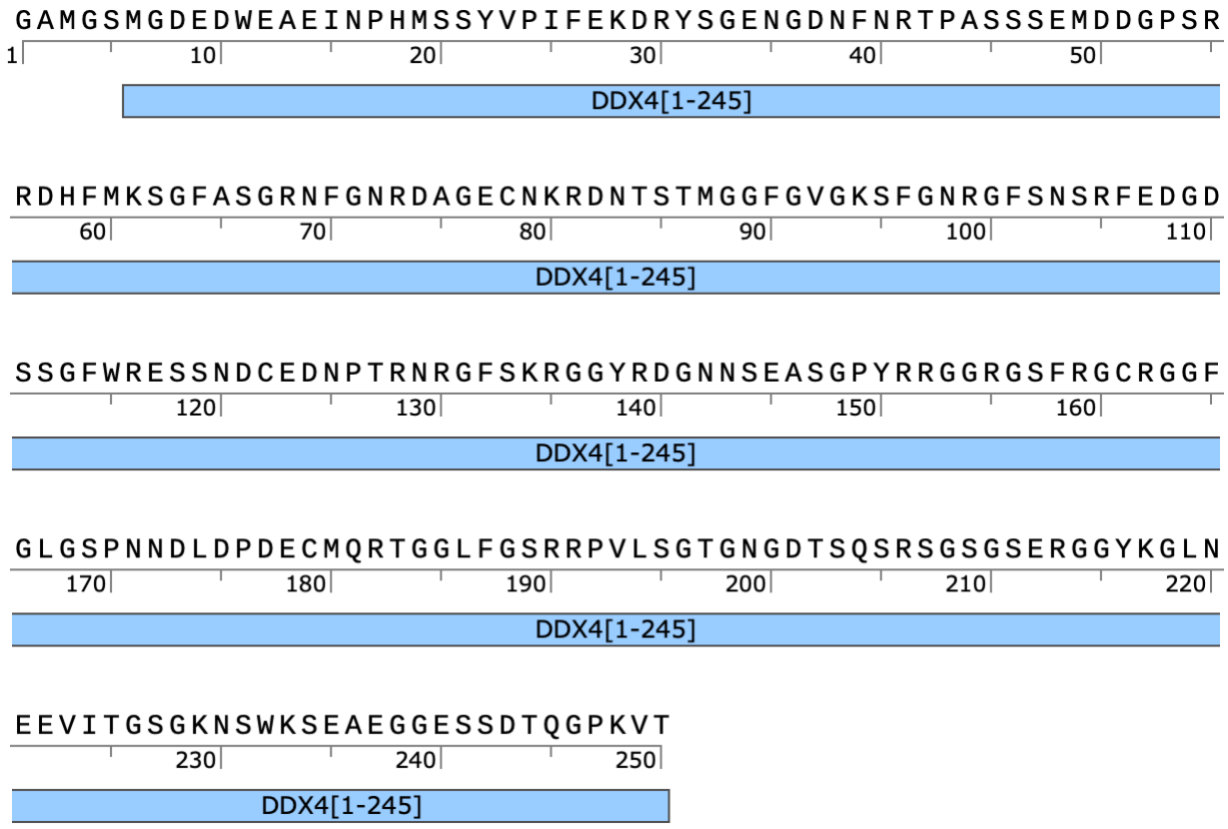
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Amino acid sequence for the experimental construct of human DDX4[1-245] used in this study, adopted from Nott, et al. (2016). (1) The first five amino acid residues (GAMGS) remain after TEV protease cleavage.

DDX4[1-245]



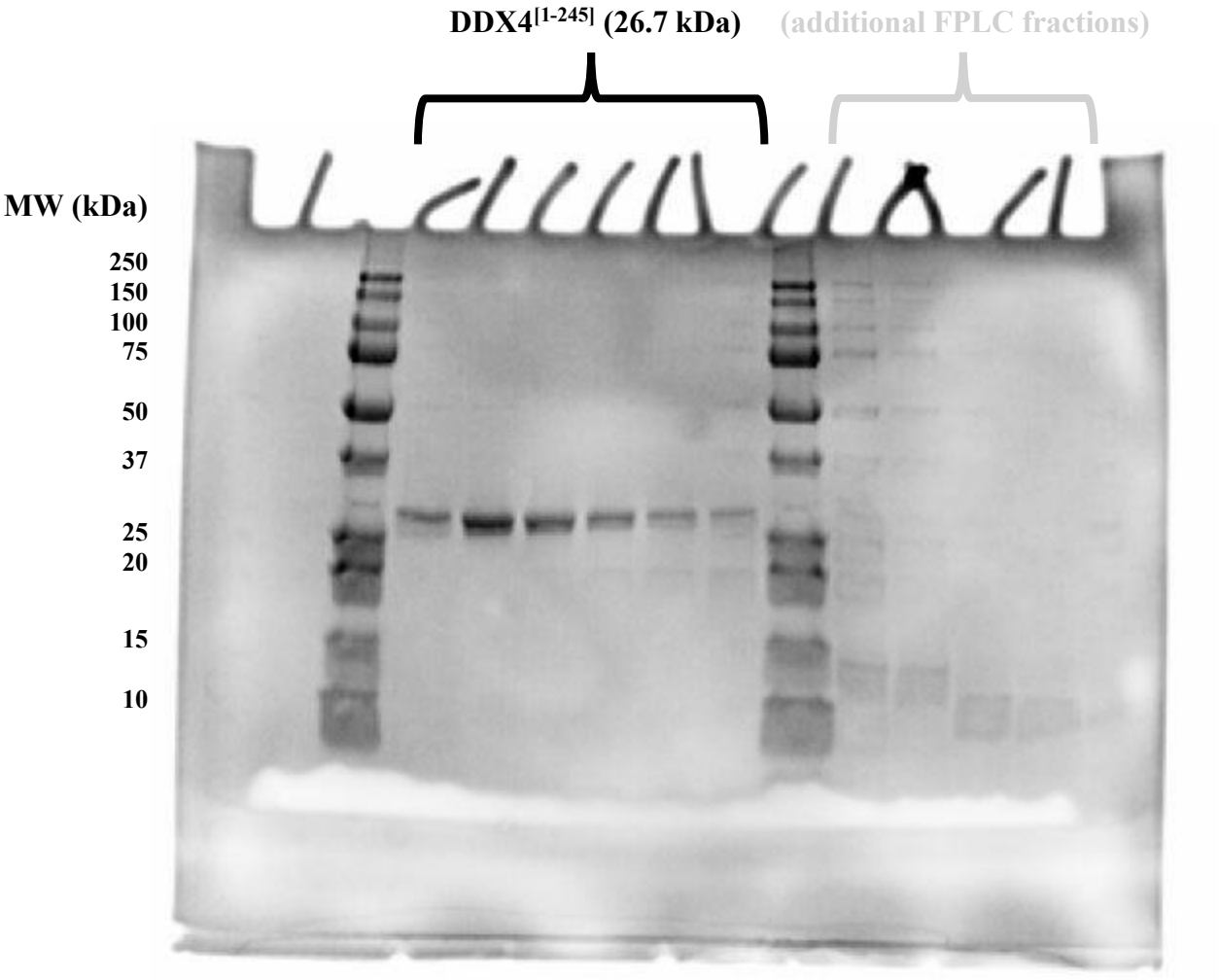


Figure S1. SDS-PAGE gel of FPLC-purified DDX4[1-245]. FPLC produces purified DDX4[1-245] used for spectroscopic analyses.

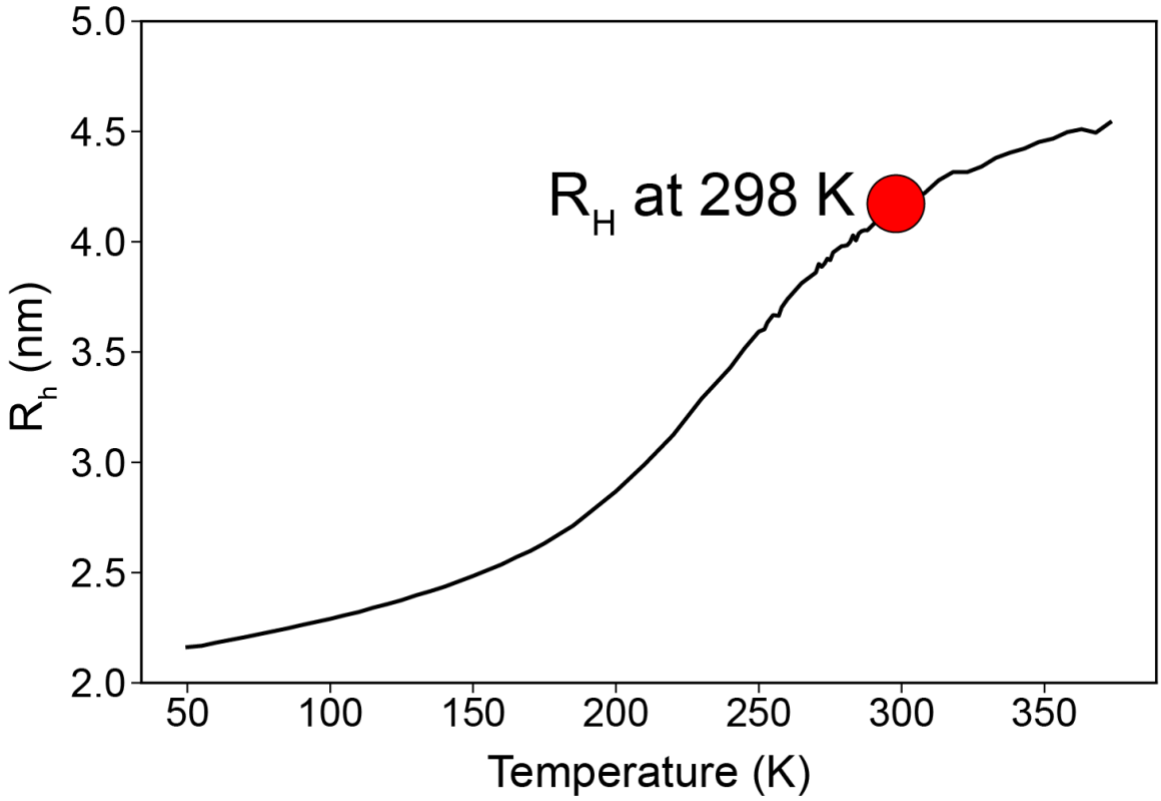


Figure S2. Theoretical coil-to-globule curve for DDX4[1-245] obtained from coarse-grained molecular dynamics simulations performed using the LAMMPS simulation engine and the Mpipi-GG forcefield. The curve here reflects a temperature-dependence in the context of the Mpipi-GG model,(2-4) but temperature should be treated here as a parameter scaling the strength of intramolecular interactions as opposed to *bona fide* temperature.

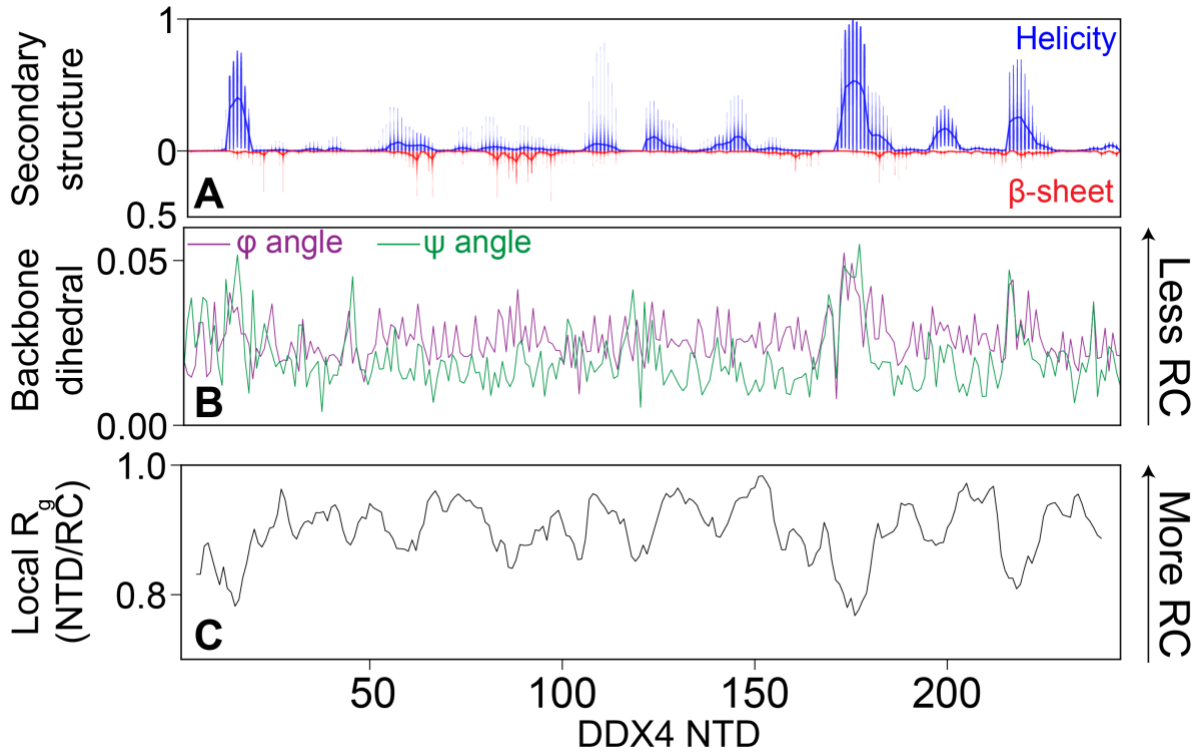


Figure S3. Assessment of local conformational properties in DDX4$^{[1-245]}$ using all-atom simulations. (A) For secondary structure assignment, the DSSP algorithm assigns local residues to either helical or 'extended' (β-strand/sheet) conformations. (B) For backbone dihedrals, the per-residue value reports on the Hellinger distance between the residue phi (purple) or psi (green) dihedral distributions with the equivalent residue's distributions from simulations performed in the Self-Avoiding Random Coil limit. Briefly, in self-avoiding random coil simulations, the same all-atom representation is used, but all attractive interactions are set to zero, leading to a conformational ensemble that recapitulates the local and global statistics of a SARC with atomistic detail. The Hellinger distance measures the overall distance between two distributions, where 0 indicates that the two are identical and 1 indicates that they are non-overlapping. The bigger the value, the less random coil (RC) like the residue is. (C) Local conformational behavior is assessed by calculating the radius of gyration for a 10-amino-acid-size "blob" within the chain and dividing by equivalent radius of gyration taken from the SARC simulation. Values closer to 1 are more RC-like, whereas values lower than 1 indicate local intramolecular interactions that lead to a more compact local radius of gyration.

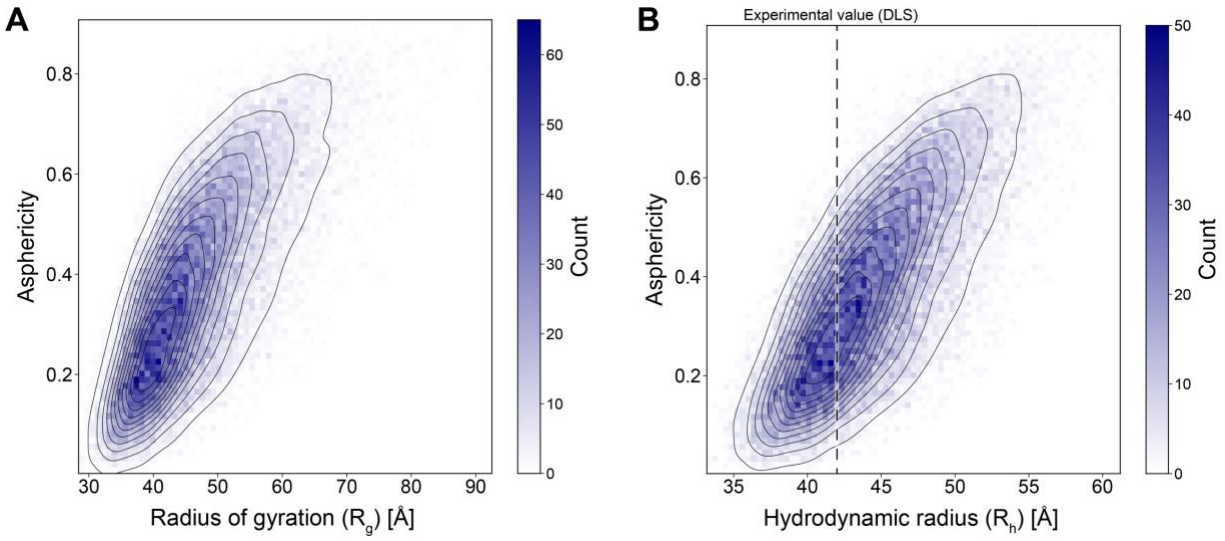


Figure S4. Assessment of global conformational properties in DDX4$^{[1-245]}$ using all-atom Monte Carlo simulations generated using the ABSINTH implicit solvent model and CAMPARI Simulation Package (ensembles contain 25,000 conformers). (A) 2D histogram (blue pixels) with Kernel Density Estimates (KDE) (black lines) overlaid for visual clarity describing the 2D distribution of the radius of gyration with asphericity. The average radius of gyration is 45.9 Å, and the broad distribution is consistent with a well-sampled highly disordered protein. (B) 2D histogram with KDE lines overlaid for visual clarity describing the 2D distribution of the hydrodynamic radius with asphericity. The average hydrodynamic radius is 44.03 Å, in reasonable agreement with the experimental value of 42 Å obtained from DLS. This broad distribution is consistent with a well-sampled, highly disordered protein.

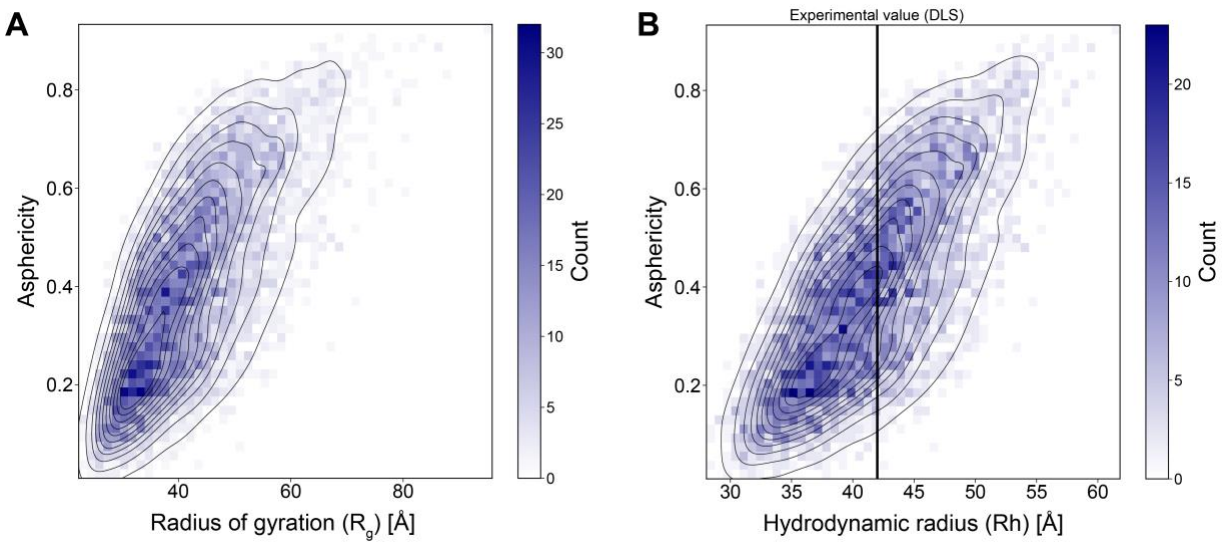


Figure S5. Assessment of global conformational properties in DDX4$^{[1-245]}$ using STARLING-derived ensembles (ensembles contain 6000 conformers). (A) 2D histogram (blue pixels) with Kernel Density Estimates (KDE) (black lines) overlaid for visual clarity describing the 2D distribution of the radius of gyration with asphericity. The average radius of gyration is 41.7 Å, and the broad distribution is consistent with a well-sampled highly disordered protein. (B) 2D histogram with KDE lines overlaid for visual clarity describing the 2D distribution of the hydrodynamic radius with asphericity. The average hydrodynamic radius is 41.4 Å, in good agreement with the experimental value of 42 Å obtained from DLS. This broad distribution is consistent with a well-sampled, highly disordered protein.

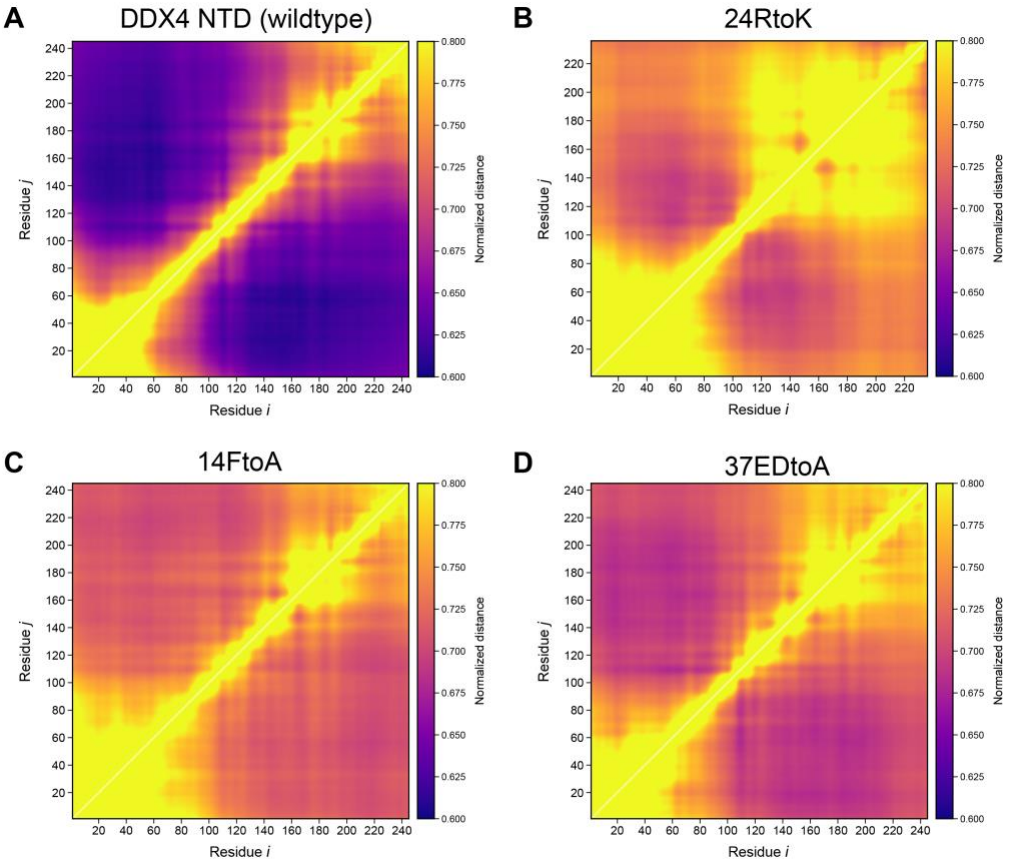


Figure S6. Intramolecular scaling maps for wildtype and DDX4 NTD variants. Scaling maps are calculated as the ratio of the distance seen in the specific variant, normalized by the distance seen for a serine-proline dipeptide repeat as a null model for an expanded, uniformly interacting IDR. (A) DDX4$^{[1\text{-}245]}$ scaling map reveals long-range interactions that deviate from an expanded chain (blue regions) across the sequence; notably, residues between 80 and 140 contribute extensive long-range interactions. These interactions are driven by a high local density of arginine-aromatic interactions, and align with subregions predicted computationally (5) and measured experimentally (6) to be critical for condensate formation. (B, C, D) A marked and overall reduction in long-range interactions is observed in the 24RtoK, 14FtoA, and 37DEtoA variants, as indicated by an increase in the normalized distances. Note the 24RtoK variant uses the DDX4$^{[1\text{-}236]}$ construct to match previously reported experimental work.

Using intrachain scaling behavior, an estimate of the apparent Flory scaling exponent (ν^{app}) for CAMPARI and STARLING derived ensembles is 0.51 and 0.52, respectively.(7) While interpreting an apparent scaling exponent has many caveats, both values are consistent with chains in marginally good solvents, close to scaling behavior expected for a polymer in a theta solvent ($\nu = 0.5$). These results are consistent with a model in which DDX4$^{[1\text{-}245]}$ is poised for environmental control of solvent quality. Indeed, the original characterization of the DDX4 NTD focused on its environmental sensitivity with respect to phase separation.**(8)** When conditions change to those that promote phase separation (decreased salt or temperature), monomer collapse is driven by the reduction in solvent quality. These considerations lead us to consider a model of DDX4 NTD phase separation where intramolecular and then intermolecular interactions are favored, first leading to monomer collapse and ultimately to phase-separated assemblies (**Figure 6**, main text).

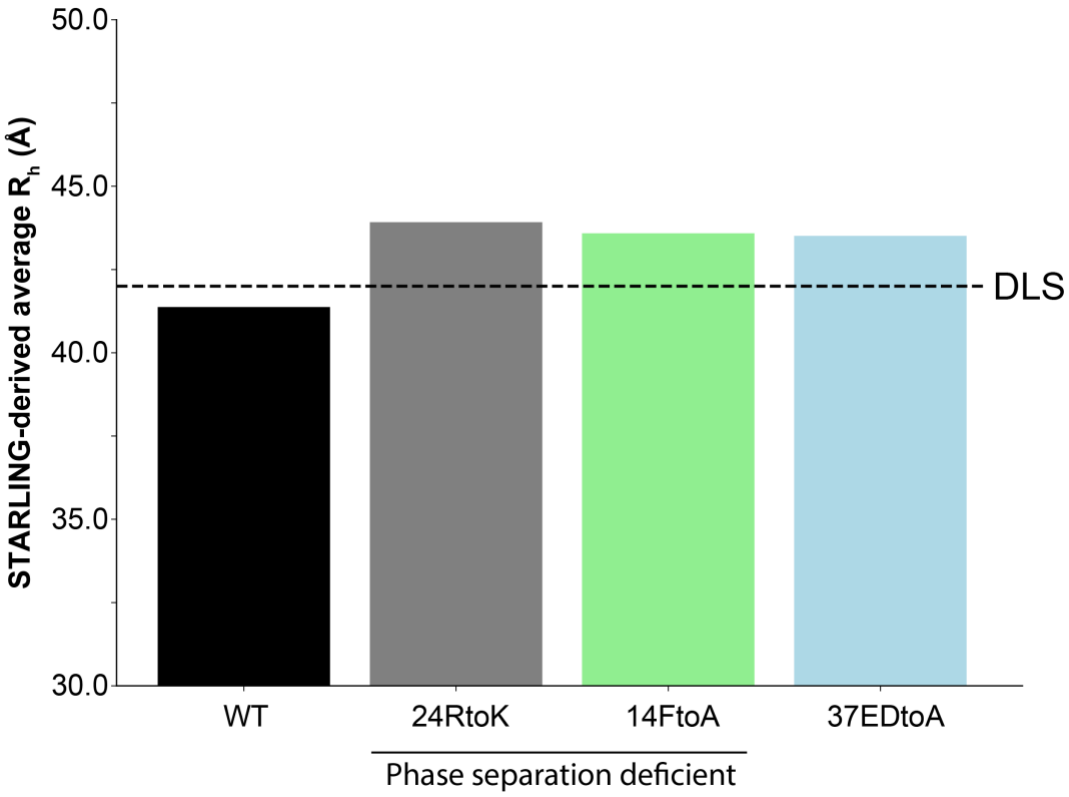


Figure S7. Predicted hydrodynamic radii from DDX4 NTD variants. In all cases, the loss of long-range intramolecular interactions (see **Figure S6**) is accompanied by an increase in overall global dimensions, as reported here by an increase in the hydrodynamic radius (R_h).

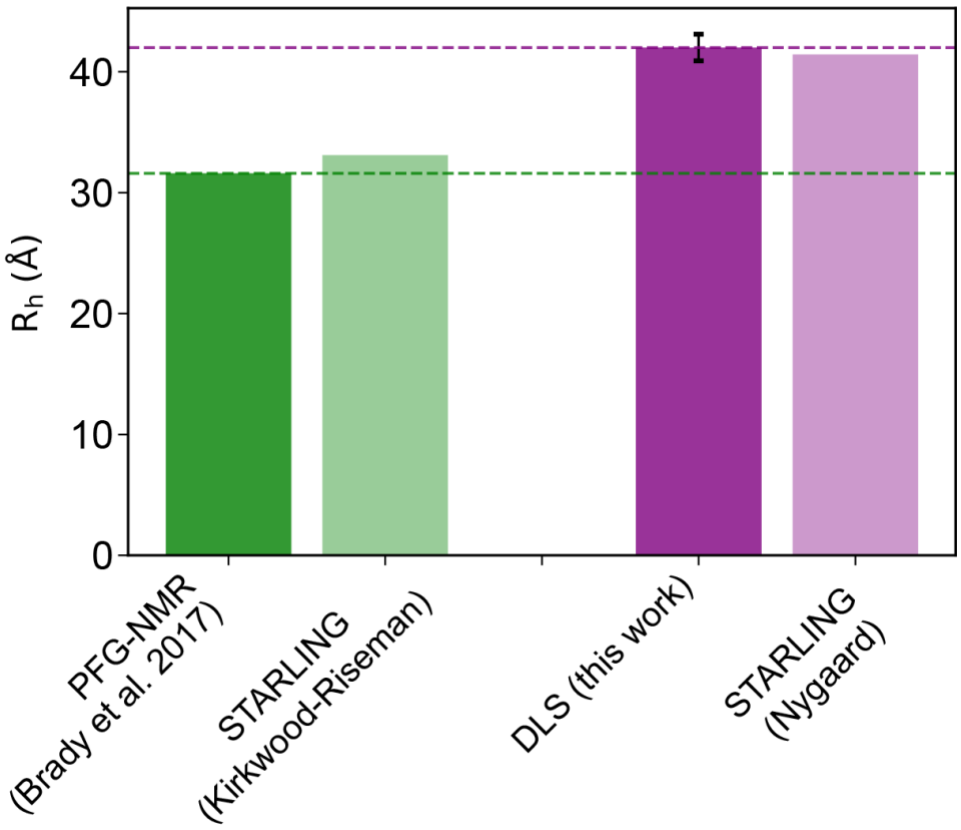


Figure S8. Comparison of experimental and STARLING-ensemble derived hydrodynamic radii (R_h) values obtained from this and prior work, using distinct methodologies. Prior work by Brady *et al.* using Pulse Field Gradient Nuclear Magnetic Resonance (PFG-NMR) spectroscopy reported a hydrodynamic radius of 3.16 nm, in extremely good agreement with the R_h derived from STARLING ensembles using the Kirkwood-Riseman approximation.(9) The Kirkwood-Riseman approximation has previously been suggested to offer better agreement with PFG-NMR-derived hydrodynamic radii, in agreement with our findings here.(10) In contrast, the R_h obtained from DLS here is, in principle, larger, at 4.2 nm; yet, for the same STARLING ensemble, an R_h of 4.14 nm is recovered using the approximation of Nygaard *et al.*,(11) which offers good agreement with our DLS data.

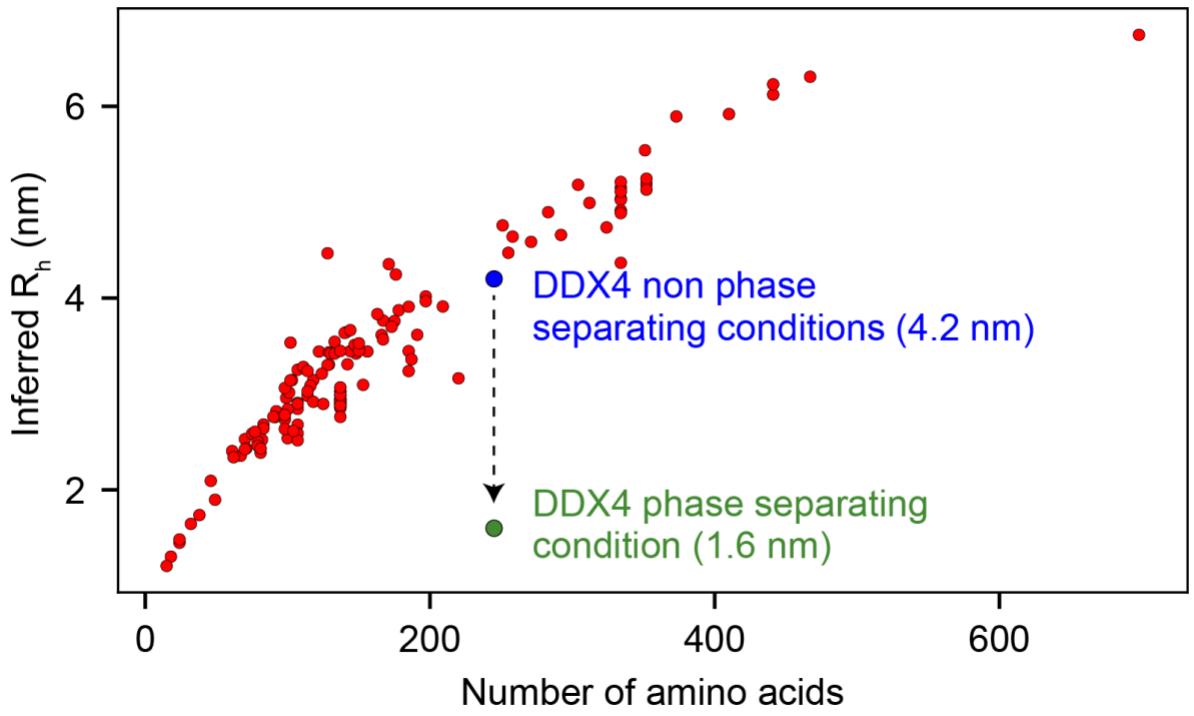


Figure S9. Comparison of inferred R_h for 144 proteins (values originally reported in Alston *et al.*).(7) R_h is calculated by taking extant R_g values obtained from SAXS data and converting R_h using the approximation of Nygaard *et al.*(11) Superimposed are our experimentally determined R_h values for DDX4$^{[1\text{-}245]}$ under non-phase separating conditions (blue) and under phase separating conditions (green) (Figure 3, main text).

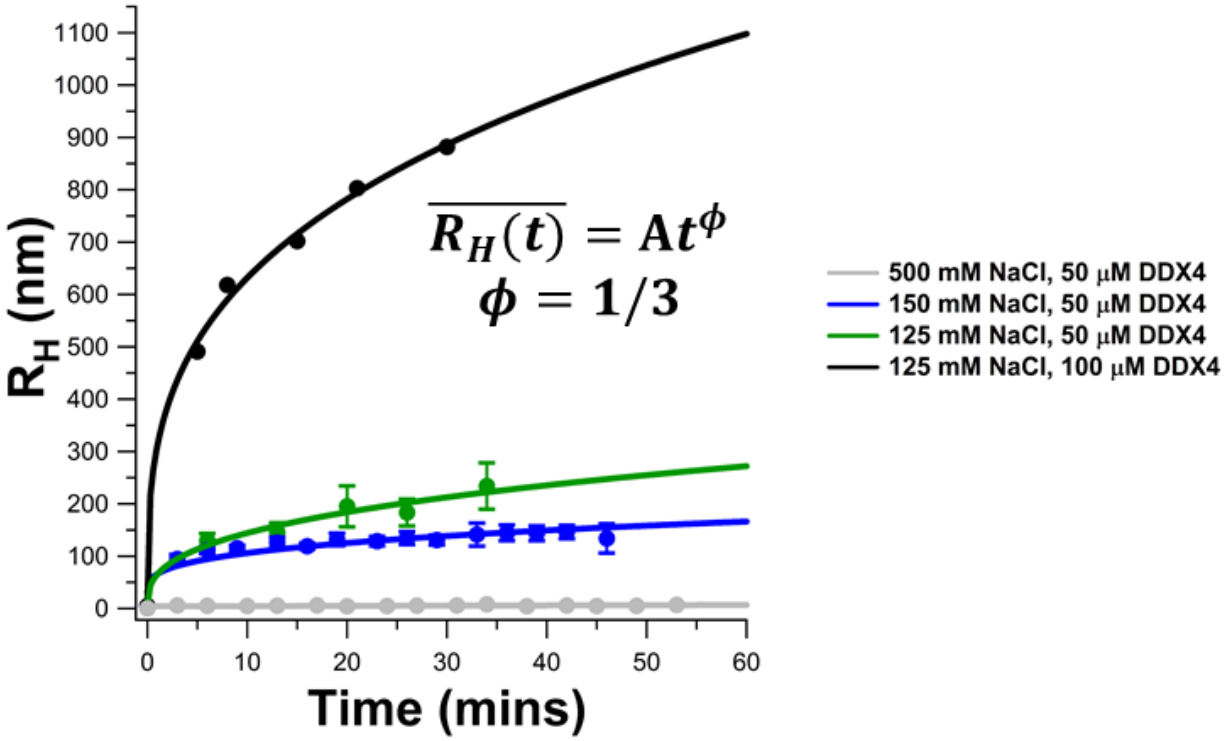


Figure S10. DLS detects growth of DDX4[1-245] condensates upon quenching to induce phase separation at 25 ºC (blue, green, black). Condensate growth curves over time follow Ostwald ripening dynamics, $< R_H(t) > = At^{(1/3)}$ (A is a pre-exponential factor). Condensates are not observed by DLS under non-phase separating conditions (gray).

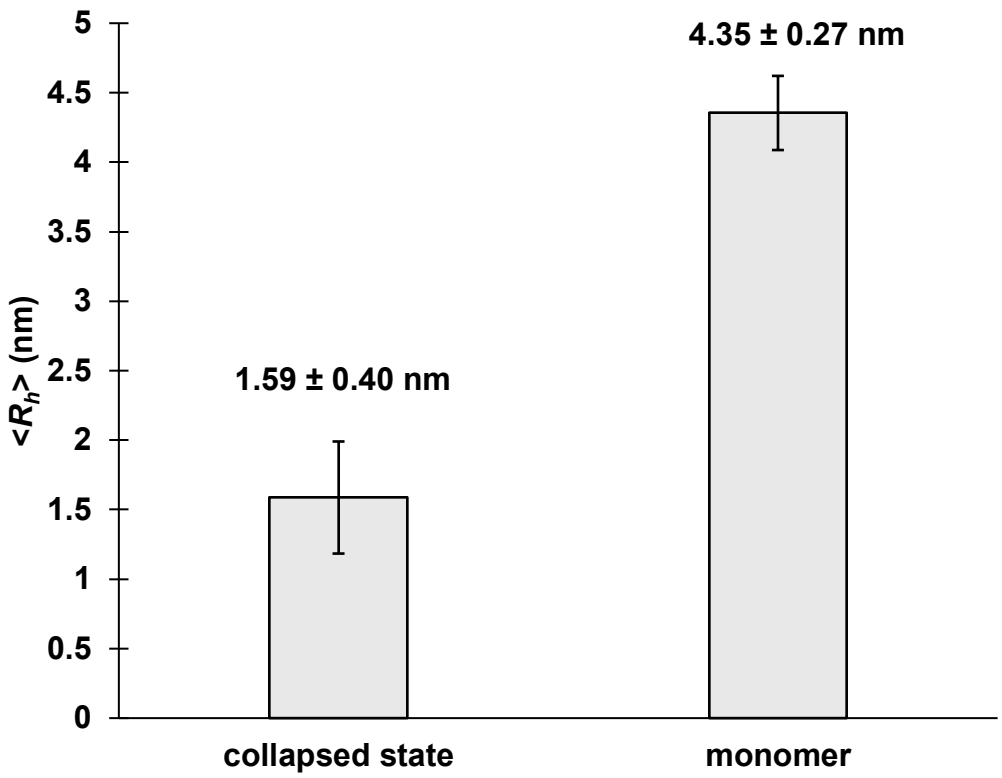


Figure S11. Dynamic light scattering reveals a reproducible collapsed state of DDX4$^{[1-245]}$ in phase separation. Average R_h from N = 11 experiments measured at 200 mM NaCl and 50 μM DDX4$^{[1-245]}$ at 10°C. Error bars indicate one standard deviation.

Table S1. UV-resonance Raman fit parameters.

Figs. 2a and 2b, all panels (solid blue curves)	**1253, 46**	**Positions, widths (cm^{-1}) for amide III/amide S disordered modes (solid blue curves)**
	1279, 23	
	1332, 51	
	1386, 30	
Fig. 2a, top panel (solid red curves)	**1230, 9**	**Positions, widths (cm^{-1}) for α-helix/β-sheet amide III modes (solid red curves)**
	1300, 36	
	1359, 35	
	1395, 9	
Fig. 2b, top panel (solid red curves)	**1230, 13**	
	1300, 36	
	1359, 36	
	1395, 9	
Figs. 2a and 2b, all panels (solid green curves)	**1163, 12-16**	**Positions, widths (cm^{-1}) for Phe modes (solid green curves)**
	1181, 17-18	
	1206, 16-20	

Table S2. Dielectric permittivity fit parameters for samples in H_2O.

	ε_s[a]	ε_∞	$\Delta\varepsilon_1$	$\Delta\varepsilon_2$	τ_2 (ps)	A_S
Water	84 ± 4	2.53 ± 0.08	77 ± 5	2.2 ± 0.5	0.3 ± 0.1	39 ± 7
Buffer	84 ± 4	2.5 ± 0.1	78 ± 5	2.1 ± 0.5	0.27 ± 0.08	36 ± 9
Dilute	78 ± 8	2.4 ± 0.2	71 ± 9	2.6 ± 0.8	0.3 ± 0.1	45 ± 7
Condensed	67 ± 6	2.58 ± 0.08	61 ± 7	2.5 ± 0.6	0.3 ± 0.1	39 ± 8

[a] The static permittivity (ε_s) is a calculated parameter.

Table S3. Dielectric permittivity fit parameters for samples in D_2O.

	ε_s[a]	ε_∞	$\Delta\varepsilon_1$	$\Delta\varepsilon_2$	τ_2 (ps)	A_S
Buffer	71 ± 4	2.6 ± 0.4	65 ± 4	2.3 ± 0.2	0.26 ± 0.03	29 ± 11
Dilute	69 ± 4	2.4 ± 0.5	63 ± 4	2.0 ± 0.3	0.27 ± 0.04	38 ± 16
Condensed	49 ± 2	2.0 ± 0.4	44 ± 3	2.1 ± 0.7	0.2 ± 0.1	41 ± 8

[a] The static permittivity (ε_s) is a calculated parameter.

Supporting Citations

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